

A Review of The Special Theory of Relativity

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Acknowledgement:

The following document is the result of an independent reading project on the topic of Special Theory of Relativity that Dr.Mathew Arun Thomas of IISER TVM asked me to start upon when I approached him.

While the conceptual understanding, explanations, pedagogical style of writing and the majority of the illustrative diagrams are **entirely my own**, the structural framework and sequence of topics in this review are heavily based on Professor Brian Greene's lectures provided by '*World Science U*'.It is worth noting that the main aim of this reading project was to cover the last 3 chapters of '*Introduction to Mechanics – by Kleppner & Kolenkow*'.

I think that the reader will definitely notice (whether he appreciates it or not) that **absolutely no AI writing tools** were used in any part of this document. Lastly, I would like to thank World Science U for making these high-quality educational resources freely accessible, which greatly facilitated my self-directed learning journey and Mathew sir whose instruction lead me to start writing this document as proof of work.

In a nutshell, this document contains my entire knowledge about the Special Theory of Relativity...

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1 Introduction

In the Spring of 1905, a storm broke out in the mind of Albert Einstein. The result of this storm was a new theory - an extension of Galilean Relativity, that revolutionized our understanding of space and time itself. In this review paper which I will use as my own notes for Special Relativity for the years to come, I shall delve deep into the intuition and mathematics of Special Relativity, derive most formulas as I go, solve example problems to strengthen the foundation and finally point the reader to the door leading to the General Theory of Relativity.

2 Why Special Relativity?

Relativity just refers to the set of rules that we have to follow when we go from one observer's frame of reference to another. It tells us what all parameters remain invariant and what all parameters vary. And if they do, by how much? The Principles of Relativity give us a general formula to convert space and time coordinates of one set of observers to the other.

2.1 Galilean Relativity

This was the first formal description of Relativity. It is what that was followed for more than 200 years. Even today, for most calculations which involve every-day speeds (much less than the speed of light), Galilean Relativity holds true to remarkable accuracy.

The postulates of Galilean Relativity are as follows :-

1. The laws of physics are identical for all inertial reference frames.
2. The length of a body is invariant (a 1 meter rod is 1 meter for all observers).
3. The time interval between two events is invariant (When 1 second passes for one observer, 1 second passes for all observers).
4. Addition/subtraction of velocities in the same direction follow algebraic addition/subtraction.

2.1.1 Addition of Velocities in Galilean Relativity

Let's say I have two observers moving in the same direction with velocities v_1 and v_2 . Their relative velocity is given by

$$v_{2,1} = v_2 - v_1$$

where $v_{2,1}$ is the velocity of observer-2 with respect to observer-1.

example: Two friends - Alex and Bob, are playing a game of catch. Alex runs towards Bob at 5m/s and throws him a ball at 10m/s. Bob who is stationary on the ground measures the speed of the ball approaching him to be 15m/s.

2.1.2 Transformations in Galilean Relativity

Let's say I have two sets of inertial observers – One is at rest and the other moving with a speed v relative to the first observer (Note that if the first observer isn't really at rest, I just choose an inertial frame moving with the specific velocity such that the observer is then at rest from that frame). Let (x, t) be the space-time coordinates of an event that happens with respect to the stationary frame. If both observers started at the same point in space at $t = 0$, then the space-time coordinates for the same event with respect to the moving observer (x', t') is given by

$$x' = x - vt \quad (\text{spatial coordinates of an event may vary})$$

$$t' = t \quad (\text{time coordinates of an event will not vary})$$

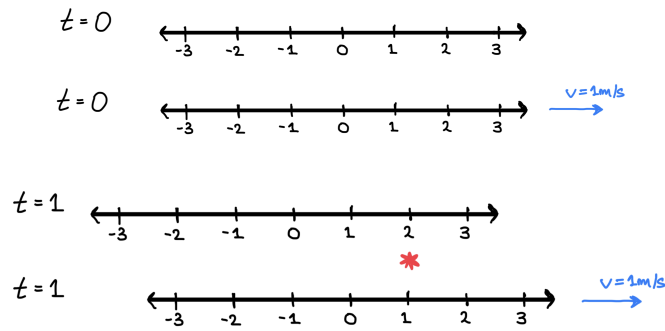


Figure 1: For the stationary frame, event(red) happens at $x = 2$. Meanwhile, for Moving frame, the same event happens at $x = 1$. Thereby verifying the Galilean transformation equation $(1\text{ m}) = (2\text{ m}) - (1\text{ m/s}) \times (1\text{ s})$.

2.2 Einstein's Relativity a.k.a Special Relativity

In the last subsection, we saw how when we change our frames from one set of observers to the other, the addition of velocities that happen, follow normal algebraic addition (for one-dimension) and normal vector addition (for 2 or 3-dimension). But there is a problem to this system. To understand the problem, we must first take a step back and ask ourselves... "What is light?"

What is Light?

Light is an electromagnetic wave which contains oscillating electric and magnetic fields that are mutually perpendicular. The oscillatory nature of both waves is crucial because according to Maxwell's Equations, only a changing magnetic field can generate an electric field and only a changing electric field can produce a magnetic field. Now imagine a sinusoidally changing electric field. It will generate a sinusoidally changing magnetic field that will again produce another sinusoidally changing electric field and so on. The fact that such an oscillation exists is the reason why light can even exist.

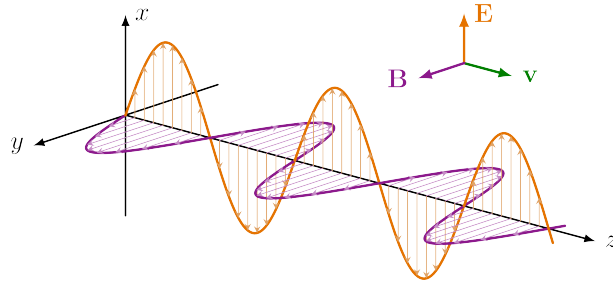


Figure 2: Light - An electromagnetic wave with mutually perpendicular $\vec{E}$ and $\vec{B}$, that are also oscillating in time.

The speed of light (or any electromagnetic wave to be precise) in vacuum – c , can be obtained purely from Maxwell's Equations as follows

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

The derivation of this doesn't specify any frame for which this speed is valid. Hence, it must be true for all frames.

Also, according to Galilean Relativity, speeds in one dimension add up algebraically. Which means, if I run alongside a beam of light at 299,792,458 m/s, I would see a section of the EM wave that is constant in space and time (ie. frozen in time). However, that leads to a paradox because in that frame, there isn't any oscillating electric field that gave rise to the magnetic field and vice versa. Hence, there can never exist a frame of reference, where the speed of light is zero (or in fact anything except c itself).

Postulates of Special Relativity

1. The laws of physics are identical for all inertial reference frames.
2. The speed of light is constant for all inertial reference frames.

That led to improvements upon the formulas of Galilean Relativity including addition of velocity and coordinate transformations

2.2.1 Addition of Velocities in Special Relativity

Let's say I have two observers moving in the same direction with velocities v_1 and v_2 . According to Special Relativity, their relative velocity is given by

$$v_{2,1} = \frac{v_2 - v_1}{1 - \frac{v_1 v_2}{c^2}}$$

(the above result is derived later)

Example Alex is holding a flashlight on a train moving at $0.5c$ and is facing the front of the train. He turns the light on and measures the speed. He measures it to be c . Bob who is standing on the ground also measures it to be equal to c . After learning Special Relativity, Alex agrees that Bob's answer is valid too. This is because, from

Alex's perspective, Bob is approaching him at $0.5c$ and the light beam is approaching Bob at $1.0c$ and the following is what he would have calculated...

$$v_{\text{light},B} = \frac{c - v_B}{1 - \frac{c \times v_B}{c^2}}$$

$$\Rightarrow v_{\text{light},B} = \frac{c - (-0.5c)}{1 - \frac{c \times (-0.5c)}{c^2}} = \frac{1.5c}{1 + 0.5} = c$$

2.2.2 Transformation in Special Relativity

Coordinate Transformations in Special Relativity, also called Lorentz Transformations is an extension of Galilean transformation. It ensures that the constant speed of light is preserved and to compensate for that it removes the condition for absolute space and time

$$x' = \gamma(x - vt)$$

$$t' = \gamma\left(t - \frac{vx}{c^2}\right)$$

Here, γ is called Lorentz factor and is given by

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

(the above results are also derived later)

3 Relativity of Simultaneity

3.1 Galilean Approach to Simultaneity

According to Galilean Relativity, events that are simultaneous for one set of observers are simultaneous for all other observers. This conclusion is drawn from the Galilean postulate that time is absolute. However relativistically, time is not absolute and therefore two events simultaneous to one set of observers need not be simultaneous to all observers. Let us look at one example where Galilean relativity seems to make sense (only because the speeds involved are very less)

Setup : Alex and Bob are on a train carriage moving forward with v_{train} . There is a two-sided ball launcher kept in the exact middle of the carriage that launches two balls to either sides towards Alex and Bob at v_{ball} . When the ball is launched, Alex and Bob measures the speed of their incoming ball to be v_{ball} and they receive it simultaneously.

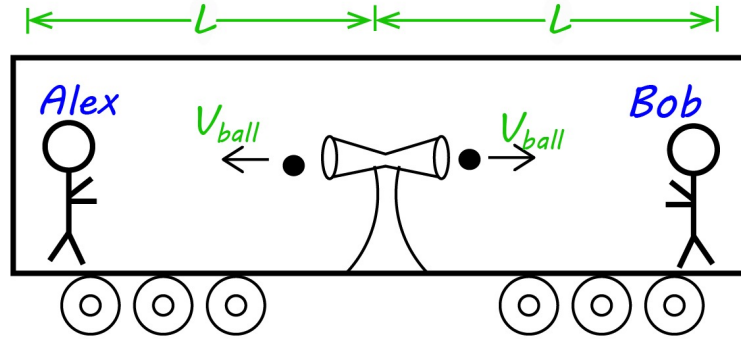


Figure 3: From the Train Perspective

But to the people on the platform, would that still be the case according to Galilean Relativity? Yes! and here's why. Suppose the length of the train carriage is $2L$ (L distance to both Bob and Alex from the ball launcher). Speed of the ball launch be v_{ball} and speed of the train be v_{train} .

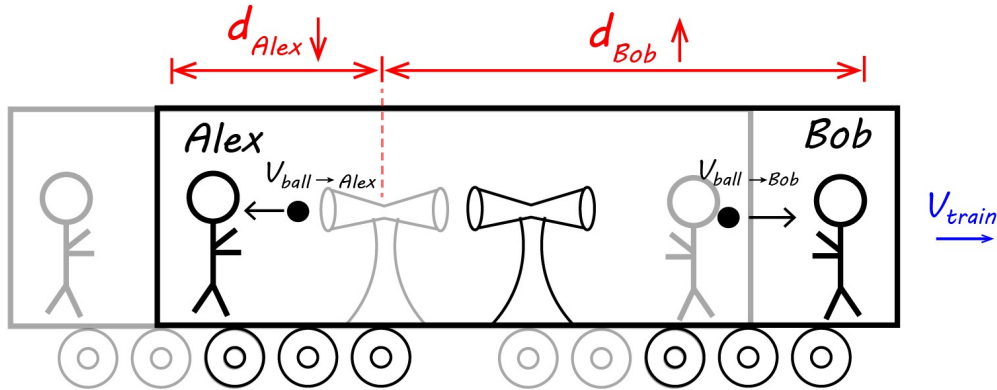


Figure 4: From the Platform Perspective. Notice how the distance traveled by the ball is more in the forward direction and less in the backward direction because the train is moving forward.

From the platform perspective, the ball launched towards Bob who is in the front of the train carriage will have an extra forward kick in velocity due to the train's forward motion.

$$v_{\text{ball} \rightarrow \text{Bob}} = v_{\text{ball}} + v_{\text{train}}$$

Similarly, the ball launched towards Alex will have its speed diminished by the same amount.

$$v_{\text{ball} \rightarrow \text{Alex}} = v_{\text{ball}} - v_{\text{train}}$$

During this time interval between the ball leaving the launcher and the friends receiving the ball, the ball must have moved forward. That is, the distance $\text{ball} \rightarrow \text{Bob} > \text{ball} \rightarrow \text{Alex}$. Since $v_{\text{ball} \rightarrow \text{Bob}} > v_{\text{ball} \rightarrow \text{Alex}}$, these two effects cancel out perfectly and the time taken by the ball to reach both the friends are identical. Not convinced? Here you go...

Let t_{Bob} be the time taken for Bob to receive his ball and t_{Alex} be the time taken for Alex to receive his. Similarly, d_{Bob} be the distance traveled by the ball (from the platform perspective) to reach Bob and similarly for d_{Alex}

$t_{\text{Alex}} = \frac{d_{\text{Alex}}}{v_{\text{ball} \rightarrow \text{Alex}}}$ $t_{\text{Alex}} = \frac{L - (v_{\text{train}} \cdot t_{\text{Alex}})}{v_{\text{ball}} - v_{\text{train}}}$ $\implies t_{\text{Alex}} (v_{\text{ball}} - v_{\text{train}} + v_{\text{train}}) = L$ $\implies t_{\text{Alex}} = \frac{L}{v_{\text{ball}}}$	$t_{\text{Bob}} = \frac{d_{\text{Bob}}}{v_{\text{ball} \rightarrow \text{Bob}}}$ $t_{\text{Bob}} = \frac{L + (v_{\text{train}} \cdot t_{\text{Bob}})}{v_{\text{ball}} + v_{\text{train}}}$ $\implies t_{\text{Bob}} (v_{\text{ball}} + v_{\text{train}} - v_{\text{train}}) = L$ $\implies t_{\text{Bob}} = \frac{L}{v_{\text{ball}}}$
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Hence, $t_{\text{Alex}} = t_{\text{Bob}}$. Notice how this is also the same time measured for the ball to reach Alex and Bob from inside the train perspective.

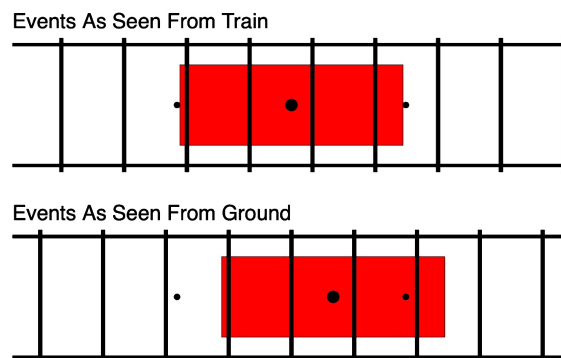
3.2 Relativistic Approach to Simultaneity

Now that we have the ‘pretty obvious’ Galilean simultaneity out of the way, let’s think more relativistically. For this, we must first top up the speeds involved to near light speed. You know what? let’s not deal with balls anymore. Lets deal with light itself.

Setup: Alex and Bob are on the same train carriage moving forward. The only difference is that now the ball launcher in the center is replaced by a light bulb that turns on exactly when the train passes the platform.

From the perspective of those on the train, since Alex and Bob are equidistant from the light bulb, the light reaches them simultaneously. No problem.

However, things are different from the point of view of those on the platform. Just as before, the light ray should travel a longer distance (compared to L) to reach Bob and a shorter distance (compared to L) to reach Alex. But unlike last time, the speed of light doesn’t get an extra kick forward or a pull backward due to the forward motion of the train since the speed of light is constant for all inertial reference frames. As there is nothing to cancel out the effect of different travel lengths, the travel time itself for light is different for reaching Alex and Bob, from the Platform perspective. What does this mean? There is a lack of simultaneity for two events in the platform frame that were simultaneous in the train frame.



Here's the same thing expressed Mathematically

$$\begin{array}{c|c}
 \begin{array}{l}
 t_{\text{Alex}} = \frac{d_{\text{Alex}}}{c} \\
 t_{\text{Alex}} = \frac{L - (v_{\text{train}} \cdot t_{\text{Alex}})}{c} \\
 \Rightarrow t_{\text{Alex}} (c + v_{\text{train}}) = L \\
 \Rightarrow t_{\text{Alex}} = \frac{L}{c + v_{\text{train}}}
 \end{array}
 &
 \begin{array}{l}
 t_{\text{Bob}} = \frac{d_{\text{Bob}}}{c} \\
 t_{\text{Bob}} = \frac{L + (v_{\text{train}} \cdot t_{\text{Bob}})}{c} \\
 \Rightarrow t_{\text{Bob}} (c - v_{\text{train}}) = L \\
 \Rightarrow t_{\text{Bob}} = \frac{L}{c - v_{\text{train}}}
 \end{array}
 \end{array}$$

$$\Delta t = t_{\text{Bob}} - t_{\text{Alex}} = \frac{L}{c - v} - \frac{L}{c + v}$$

$$\Delta t = \frac{2L v}{c^2 - v^2}$$

Where, $2L$ is the length of the train from the ground perspective.

Hence the time interval for the light to reach Alex is different from that of Bob. Notice how when $v_{\text{train}} \ll c$, the effect is negligible and both $t_{\text{Bob}} = t_{\text{Alex}} \approx \frac{L}{c}$ which agrees with Galilean Relativity. This is the key idea, Galilean Relativity isn't wrong per se. It's just inaccurate for speeds close the speed of light. Special Relativity takes everything into account giving exact answers both at low speeds and high speeds for the price of cumbersome calculations.

4 Time in Motion

4.1 Why should Time be affected?

In the previous section, we found that observers in relative motion do not agree on which events happen simultaneously. To test this, suppose we place pre-synchronized clocks at both ends of the train. People on the platform will read these moving clocks as the light passes Alex and Bob, attempting to convince the train observers that the events were non-simultaneous.

We might expect this to lead to a definitive disparity in opinion. If a platform observer sees the light hit Alex and Bob at different times as per clocks on the train, the train observers would seemingly have to accept this non-simultaneity as reality.

Interestingly enough, this doesn't happen. Because when the platform observer takes his measurements using the clocks on the train, this is what he sees: the light ray hits Alex first, but that clock is "ahead" and reads a future time compared to the one in the middle. He waits for the light to pass Bob and observes a similar discrepancy; this time, the clock is "behind" and reads the exact same time that was on Alex's clock earlier when the light passed him. To the platform observer, the mystery is solved—the train observers only think the events are simultaneous because their clocks are out of sync! He waits for the train to stop to put his case forward, only to be surprised: the clocks are now perfectly synchronized. The only difference is that both train clocks read a lesser time than the stationary platform clocks, despite everyone starting the journey in sync.

The conclusion? **Clocks in motion tick-off time differently than the ones in rest. Time itself moves differently in a moving frame compared to a stationary frame.**

4.2 How fast does time slow?

For this subsection, I am going to introduce you to a new kind of clock called a light clock. It works as follows. It is a cubical box with its top and bottom surfaces being parallel mirrors facing inwards. A photon of light bounces up and down between these mirrors and corresponds to Tick-Tock of a clock. It is important to note that conceptually, the light clock is no different from an ordinary clock. Any conclusion about the nature of time that we arrive using a light clock is valid for all clocks.

In a frame where the light clock is at rest, a single tick of the tick-tock takes a time of $\frac{d}{c}$. However, when the light clock moves, the photon must travel along a diagonal path. Because special relativity dictates that the speed of light remains constant at c , the photon has to cover a longer distance, $L = \frac{d}{\sin \theta}$ at the same speed c , where θ is the angle between the path of the light and the mirror surface.

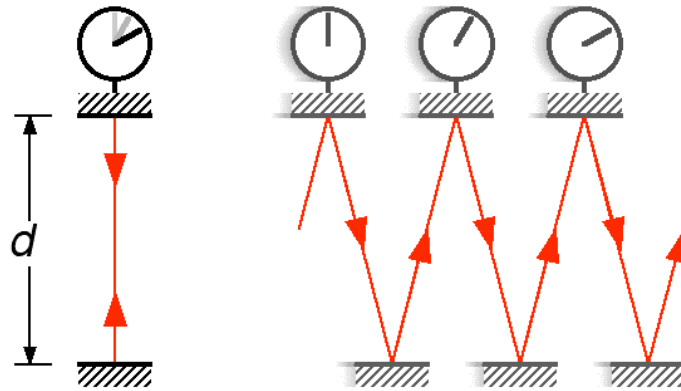


Figure 5: Light Clock in Rest and in Motion

Let's first think about what we expect to happen before deriving it mathematically. If the light clock is moving, the photon has to travel further at the same speed for each tick-tock. Hence each tick-tock in the moving clock takes longer time from the stationary frame. *Longer tick-tocks mean less time has elapsed* on the moving clock than if it were to be at rest.

Say the light clock is set in motion, moving with a speed v rightwards. A stationary observer on the ground observes the diagonal path taken by the photon to hit each mirror. For him, each tick takes L/c time on his clock to complete.

He also notes that, if a person were to move with the light-clock, the photon would just move in a simple up-and-down fashion and therefore such an observer would say that each tick would take only d/c time to complete.

Conceptually, this means that when d/c time passes for the moving observer, L/c time passes for the stationary observer. Lets call the former as t and latter as t' . Then...

$$t' = \frac{L}{c}$$

$$t' = \frac{1}{c} \left(\frac{d}{\sin(\theta)} \right) = \frac{1}{c} \left(\frac{d}{\sqrt{1 - \cos^2(\theta)}} \right)$$

Now, since θ is the angle made by the light ray with the mirror surface, $\cos(\theta)$ which is the adjacent side to the hypotenuse, is given by

$$\cos(\theta) = \frac{\text{speed of horizontal movement of light}}{\text{speed of diagonal movement of light}} = \frac{v}{c}$$

substituting for $\cos^2(\theta)$ in the earlier expression, I get...

$$t' = \frac{1}{c} \left(\frac{d}{\sqrt{1 - \frac{v^2}{c^2}}} \right)$$

I know that $\left(\frac{d}{c}\right)$ is simply the time elapsed for the moving observer – t . Hence I obtain...

$$t' = \frac{t}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Its hard to miss Lorentz factor on the right hand side of the equation. As discussed before, $\gamma \geq 1$. Hence, time elapsed on a stationary clock is always more than that elapsed on a moving clock. This phenomenon is called ‘Time Dilation’

Time Dilation Example - Space Travel

We have a spaceship headed towards *Proxima Centauri* which is 4.25 light-years away at $0.8c$. How long does this journey take with respect to observers on earth and for observers in spaceship?

Answer : Let t be the time it takes for the journey with respect to observers in the spaceship and t' be the time it takes with respect to observers on earth. For people of earth, the calculation is fairly straightforward

$$t' = \frac{d}{v} = \frac{4.25 \text{ [light-years]}}{0.8 \left[\frac{\text{light-years}}{\text{year}} \right]}$$

$$\implies t' = 5.3125 \text{ years}$$

However, for people on the spaceship, the journey time is reduced by a factor of γ . Hence

$$t = \frac{t'}{\gamma} = t' \sqrt{1 - \frac{v^2}{c^2}}$$

$$\implies t = 5.3125 \times \sqrt{1 - 0.8^2} = 3.1875 \text{ years}$$

Therefore, a journey of 5.3125 years actually only takes 3.1875 years from the space-ship perspective.

Definition - Proper time (τ)

The proper time between two events is the time that elapses on a clock that moves through both the events and for which both events take place at the same point in space. In simpler terms, it is the time that elapses on frame that is at rest relative to the object under consideration. In most cases, this is the time that reads on the clock carried by the object itself as it moves through space-time.

Time Dilation Example - Muon Decay

Muons are subatomic particles that come under leptons. It is very similar to an electron but is quite heavier than one. The interesting thing about muons is that they are unstable and they quickly disintegrate into other electrons and neutrinos. The lifetime of a muon has been measured to be 2.2 microseconds. When Muons are produced in the upper atmosphere, they rush towards the earth surface but they decay after traveling some distance. Scientists measured that the distance the muon travels in the upper atmosphere is around 10 km. However, to travel this much distance, it should have been traveling at well over the speed of light or it has not decayed well past its decay time. How would you resolve this seeming paradox?

Answer : The key idea here is that the lifetime of the muon is measured from the Muon's perspective itself. Let $\Delta\tau = 2.2\mu\text{s}$ be the decay time of the muon in proper time. Since the muon is moving, a time $\Delta\tau'$, more than $2.2\mu\text{s}$ will pass from our perspective before the Muon decays. It is like if the muon is holding a gun against itself and it will pull the trigger when the time reads $2.2\mu\text{s}$ on its clock. Due to Time Dilation, even when our clock has struck $2.2\mu\text{s}$, the muon will still have not pulled the trigger because its clock hasn't yet reached $2.2\mu\text{s}$. This is why the muon seems to live longer than it should. Calculating the speed,

$$\begin{aligned} d &= v \times (\Delta\tau') = v \times (\gamma\Delta\tau) \\ d &= v \times \frac{\Delta\tau}{\sqrt{1 - \frac{v^2}{c^2}}} \\ \Rightarrow 10^4 &= v \times \frac{2.2 \times 10^{-6}}{\sqrt{1 - \frac{v^2}{c^2}}} \end{aligned}$$

Solving for v , we get, $v = 0.9978c$. Hence, for a muon which travels at 99.78% the speed of light, it can travel about 10km before decaying, while still traveling at a speed less than the speed of light. Although it first looked like the muon was living more than it should, now we know that it does not.

5 Motion's Effect on Space

5.1 Why should Space be affected?

In the previous section we saw how time is not absolute. The time interval between two events depends upon the relative velocity of the observers. However, since speed of light is constant, and $c = \text{distance traveled} / \text{duration of travel} = \text{const.}$, the non-absolute nature of time would imply a non-absolute nature of space so that their ratio would remain constant. Let me guide you through a thought experiment

setup: Two friends - Reyna and Soundarya decide to measure the speed of light by sending out a laser pulse on the ground towards a detector 1 km away, measure the time it takes for laser pulse to reach the detector and hence calculate the speed of light. They both have learnt Special Relativity and decide to use this as a good opportunity to test the constant nature of the speed of light. For this, Reyna measures the speed of light from the ground (stationary) perspective meanwhile Soundarya measures the speed of light from inside of a moving train using her own stopwatch. Reyna is aware that due to the Soundarya's motion, time should tick off more slowly for her and therefore Soundarya should see a lesser time elapsed on her stopwatch for the journey of the light beam towards the detector. Thus Reyna predicts that the speed that Soundarya would calculate for the laser pulse – c' should be more than c because

$$c' = \frac{1 \text{ km}}{t_{\text{Soundarya}}}$$

$$t_{\text{Soundarya}} < t_{\text{Reyna}} \implies c' > c$$

To Reyna's surprise, when she compares her measured speed to that of Soundarya, she sees that both are equal! How does Reyna come up with a reasoning as to how Soundarya could arrive at the same speed, despite having different travel time for the laser? The secret lies in the distance traveled by the laser from Soundarya's perspective. Yes it is true that the duration of light's travel time has shrunk for Soundarya, but so has the distance traveled by the light from her perspective. These two effects cancel out perfectly and we arrive at the same speed. Hence $c' = c$. This space-shrinking phenomenon is called '*Length Contraction*'.

The conclusion? **Rods in motion shrink in the direction of its motion from the perspective of someone stationary. It is important to note that height and width remain the same as long as there is no component of velocity in those directions.**

5.2 Length Contraction as a consequence of Time Dilation

The above example would have hopefully given you an idea for why length contraction is essential to keep the speed of light constant. But I still haven't showed you the physical 'length' of a moving object shrinking as a consequence of Special Relativity. More importantly, I will show you how length contraction effects from one frame is a result of time dilation effects on another. Buckle up for another thought experiment.

setup: Two friends – Alice and Bob, decide to measure the length of a moving train two different ways. Bob will stay inside the train and measure the distance by moving a tape measure from the back of the train till the front. Meanwhile Alice who is on the ground has a different approach. She has marked a spot on the ground. When the front of the train passes that spot, she starts her stopwatch and when the rear of the train passes it, she stops her stopwatch. She multiplies the speed of the train with the duration of travel to come up with the length of the train.

$$\text{Length of train} = \text{speed} \times \text{duration}$$

However, when they do measure the length this way, they arrive at very different answers. Bob got, say 240 m. While Alice got a shorter length, say 180 m. This is length contraction in action. But does this puzzle Bob who understands time dilation? No. Here's why

From Bob's Perspective, Alice is in motion and hence her clocks are ticking off time more slowly. Therefore, Alice would have measured a smaller time interval between the front of the train and rear of the train passing the fixed point on the ground. When she multiplies that smaller time with the speed of the train, she gets a smaller length.

5.3 Mathematical Form of Length Contraction

In the earlier example, we saw how Alice could measure the length of the train by multiplying the velocity of the train with the time interval–(t_A) between the front and rear of the train passing a fixed point on the ground. Bob could play the same game but with two clocks instead of one – One at the front end of the train and other at the rear end (both clocks already started in sync with respect to the train frame). When Alice passes the front of the train, Bob stops the clock in the front end and when Alice passes the end of the train, Bob stops the clock at the rear end. He then takes the time difference–(t_B) between the readings on the rear clock and front clock. Let L_A and L_B be the length of the train as measured by Alice and Bob respectively.

$$L_A = v \times t_A \quad \Bigg| \quad L_B = v \times t_B$$

From Bob's Perspective, Alice is in motion and her clocks are ticking off time more slowly. (Note that the reason why we can't apply time dilation to Bob's frame from Alice's perspective is because when the train is passing her, she notices that the clock at rear of the train is already ahead in time compared to the front of the train and hence concludes that Bob would anyway get an answer bigger than what he should have)

$$t_A = \frac{t_B}{\gamma} = t_B \sqrt{1 - \frac{v^2}{c^2}}$$

substituting this for t_A , we get

$$L_A = v \times \frac{t_B}{\gamma} = \frac{L_B}{\gamma}$$

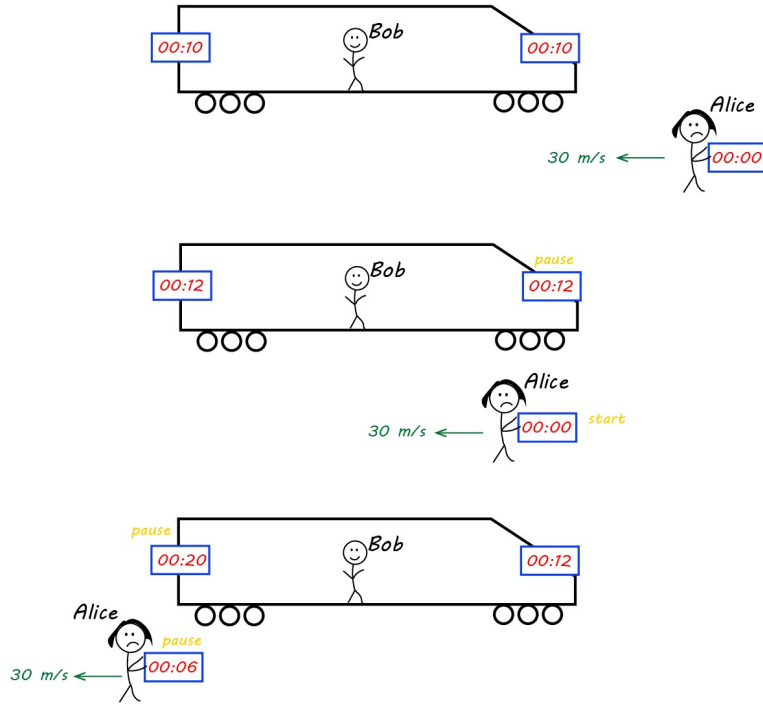


Figure 6: Events from Bob's perspective. Alice is in motion and hence her stopwatch should tick off time more slowly. He concludes that this is why Alice got only 6 seconds on her clock meanwhile he himself measures a time difference of 8 seconds between his clocks. $L_B = 30 \text{ m/sec} \times 8 \text{ sec} = 240 \text{ m}$

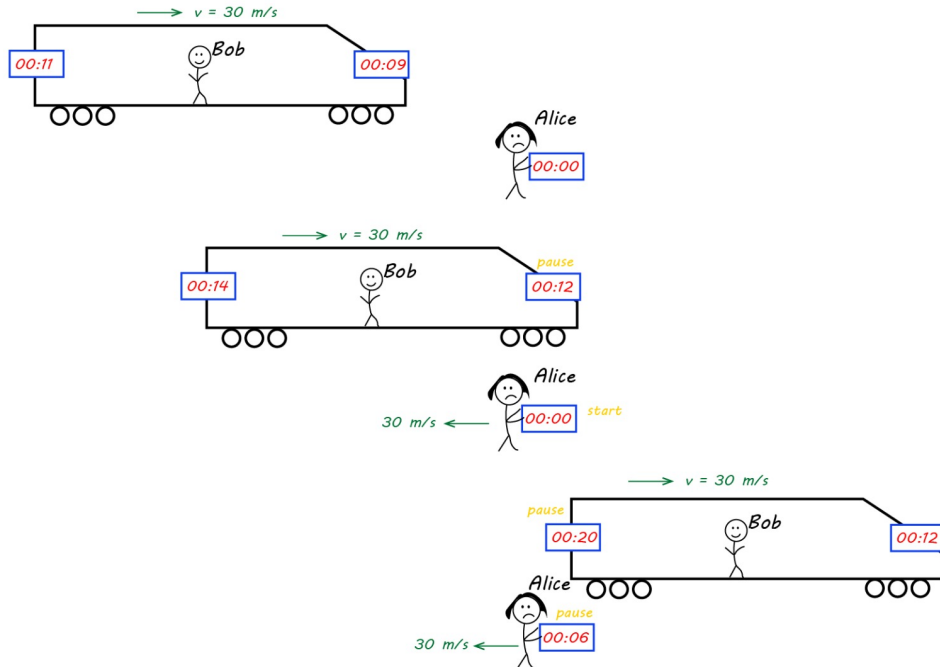


Figure 7: Events from Alice's Perspective. Two things come into play here. First of all, since Bob is in motion, all of his clocks tick off time more slowly. Secondly, Due to his forward motion, clocks at the rear are ahead in time and those at the front are behind in time. There is a 2 second difference between the clocks in the front and the rear. She concludes that this is why Bob measures an additional 2 seconds. $L_A = 30 \text{ m/sec} \times 6 \text{ sec} = 180 \text{ m}$

Therefore, the length of a moving object – L , is measured from a stationary frame to be L' , such that

$$L' = \frac{L}{\gamma} = L\sqrt{1 - \frac{v^2}{c^2}}$$

Role of Simultaneity for Measuring Lengths

In the above example, we had Alice wait for sometime and thereby measure the time interval between two events for the purpose of measuring the length of the train. However, what if we could find a work-around by not having any time interval that undergoes time dilation? In other words, we measure the position of the front and back of the train simultaneously and then measure the distance between these points. Interestingly enough, then too we arrive at the exact same result. Reason? Lack of simultaneity between moving observers! Although Alice may claim that the positions of front and back of the train at a single instant was taken simultaneously, Bob would disagree because in his frame, Alice and the earth is moving backwards and the two events would have occurred one after the other. It would seem as if the position of the front of the train was measured first, the train moves a bit further, then the position of the rear of the train is measured. Hence, arriving at a shorter length once again.

Length Contraction Example - Space Travel

We have a spaceship headed towards Proxima Centauri which is 4.25 light-years away, at $0.8c$. How long does this journey take with respect to observers on earth and for observers in the spaceship?

Answer : From the point of view of the observers on the spaceship, the distance between Earth and Proxima Centauri gets Lorentz Contracted by a factor of γ .

$$d_{\text{spaceship}} = \frac{d_{\text{earth}}}{\gamma} = 4.25 \text{ [light-years]} \times \sqrt{1 - 0.8^2} = 2.55 \text{ light-years}$$

And the hence, the time it takes for this journey becomes

$$t = \frac{2.55 \text{ [light-years]}}{0.8 \left[\frac{\text{light-years}}{\text{years}} \right]} = 3.1875 \text{ years}$$

Notice how this is the same answer we got earlier but then by using Time Dilation in the spaceship from earth observers. Meanwhile, here we utilized length contraction outside the spaceship from the frame of observers inside the spaceship. This example confirms once again that Time-Dilation in one frame comes out as Length-Contraction in another.

Length Contraction Example - Muon Decay

Previously, we had seen how the paradox of a muon traveling longer than it should could be solved by using Time dilation of the muon from our perspective. Now we'll see that there is a far easier way to tackle this problem by thinking of Lorentz Contraction. Calculate the time for which the muon stays alive in our frame, in its frame and the distance it moves through space in our frame and its frame. Remember, decay (proper) time of muon is $2.2 \mu\text{s}$ and the muon appears to travel 10 km in the upper atmosphere before it decays.

Answer: From the Muon's perspective, it is stationary and the rest of the world is rushing past it at v . Hence the distance it travels become Lorentz contracted by a factor of γ .

$$d_{\text{Muon}} = \frac{d_{\text{earth}}}{\gamma} = 10^4 \sqrt{1 - \frac{v^2}{c^2}}$$

Now, from the Muon's perspective, it travels a distance of d_{Muon} by the time $2.2 \mu\text{s}$ passes on its clock. If the speed of the Muon is v ,

$$\begin{aligned} d_{\text{Muon}} &= v \times 2.2 \times 10^{-6} \text{ [sec]} \\ \Rightarrow 10^4 \sqrt{1 - \frac{v^2}{c^2}} &= v \times 2.2 \times 10^{-6} \text{ [sec]} \end{aligned}$$

Solving for v , we get $v = 99.78\%$ speed of light as before. This gives us a lorentz factor - $\gamma = 15.08$.

Plugging this into Length contraction and time dilation formulas, we get

$$L' = \frac{10 \text{ km}}{\gamma} = 663 \text{ m}$$

and

$$t' = \gamma \times 2.2 \text{ microsecond} = 33.17 \text{ microseconds}$$

Although it seems from our perspective that the muon traveled 10 km and didn't decay for 33.17 μs , From the Muon's frame, it has traveled only a total distance of 663 m and has stayed alive without decaying only for 2.2 μs .

Length Contraction Example - Length Contraction with Relativity of Simultaneity

We revisit our earlier example on Alex and Bob at two ends of a train with a light bulb in the middle. They both claim that they are 5 ft from the light bulb. At what speed should the train be moving so that the platform observers see Alex receiving the light beam 1 nanosecond before Bob (use approximation : $c \approx 1\text{ft}/\text{nanosecond}$)

Answer: Remember our earlier formula to calculate the time difference between simultaneous events as observed by someone on the platform

$$\Delta t = \frac{2Lv}{c^2 - v^2}$$

However, just as I mentioned before, $2L$ here refers to the length of the train from the platform perspective and is not $2 \times 5 \text{ ft} = 10 \text{ ft}$. The length would be Lorentz Contracted by a factor of γ .

$$\Delta t = \frac{2L' v \sqrt{1 - v^2/c^2}}{c^2 - v^2}$$

where $2L'$ is the proper length of the train = 10 ft

$$\begin{aligned} 1 \text{ ns} &= \frac{10 \text{ [ft]} v \sqrt{1 - v^2/1^2}}{1^2 - v^2} \\ 1 \text{ ns} &= \frac{10 \text{ [ft]} v \sqrt{1^2 - v^2}}{(1^2 - v^2)} = \frac{10 \text{ [ft]} v}{\sqrt{1^2 - v^2}} \\ \Rightarrow v &= \sqrt{\frac{1}{101}} \text{ ft/ns} \end{aligned}$$

Hence, this happens when $v \approx 0.1c$ or 10% the speed of light.

With this, I bring both Time Dilation and Length Contraction to a conclusion. Below given are the key takeaways and points to remember while solving a relativistic problem involving either one of these effects.

Things to take care when solving Time-Dilation and Length-Contraction problems.

1. Before dealing with Time Dilation or Length contraction, always be aware of from whose frame is the effect observed and on what is the effect observed. That is, form the following sentence in your head – “From the frame of reference of co-ordinate system-1, coordinate system-2 is length contracted (or experiences time dilation).” Remember that a system can never experience length contraction or time dilation on itself.
2. Relativistic effects never affect the agreement of relative velocity between two observers. If observer-1 sees observer-2 approaching it as $0.95c$, observer-2 will necessarily also see observer-1 approaching him at the same speed. However, they will almost certainly disagree on distances and the time interval between events.

6 Coordinates in Motion

So far, we saw that observers in relative motion do not agree on space, time and simultaneity. What we want now is one mathematical framework that will put all of this together. For this, we need to first start thinking of reality as a collection of events that happen at particular point in space and a particular point in time.

The crux of Special Relativity is to answer the following questions

1. “Where and when do **I** say the sequence of events take place?”
2. “Where and when do **someone else** say that the sequence of events take place?”

6.1 Coordinates of Space

Pretty much always we use the Cartesian coordinate systems for space. Ideally 3-dimensional but 1-dimensional for simplicity. It is important to emphasize here that there is no absolute correct answer for position for an event in space. It all depends upon the coordinate system we choose. Different coordinate systems in space only differ in two ways – Translation and Rotation.

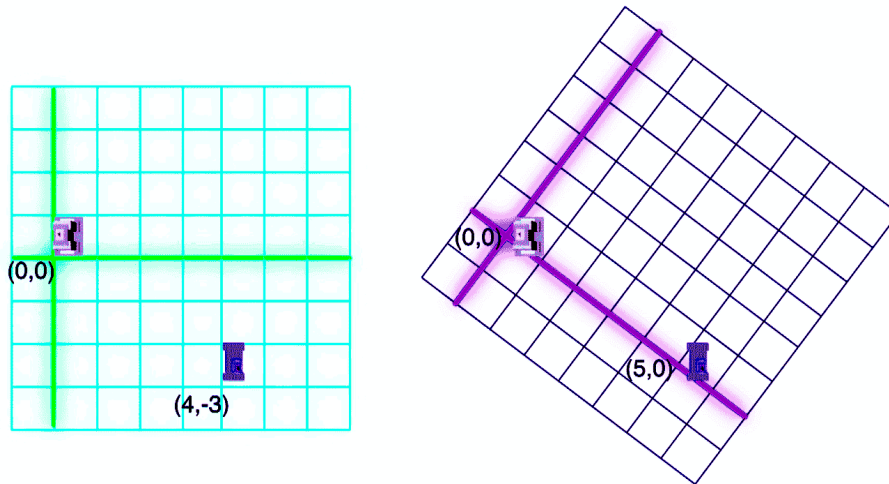


Figure 8: Two Coordinate systems having the same origin showing different positions for two buildings because of rotation

To distinguish between coordinate systems. I use the primed and unprimed notation I followed till now. When I say x is the position of an event in the unprimed reference frame, what I mean is – the position that me as an observer would put for an event if I was standing on the origin of the unprimed reference frame (and moving with it if it is). Similarly for x' being the position of an event in the primed reference frame with respect to an observer observing from the origin of this primed frame.

Let us now go through a brief example of how to convert the position coordinates of an event from one coordinate system to the other in the case of rotation.

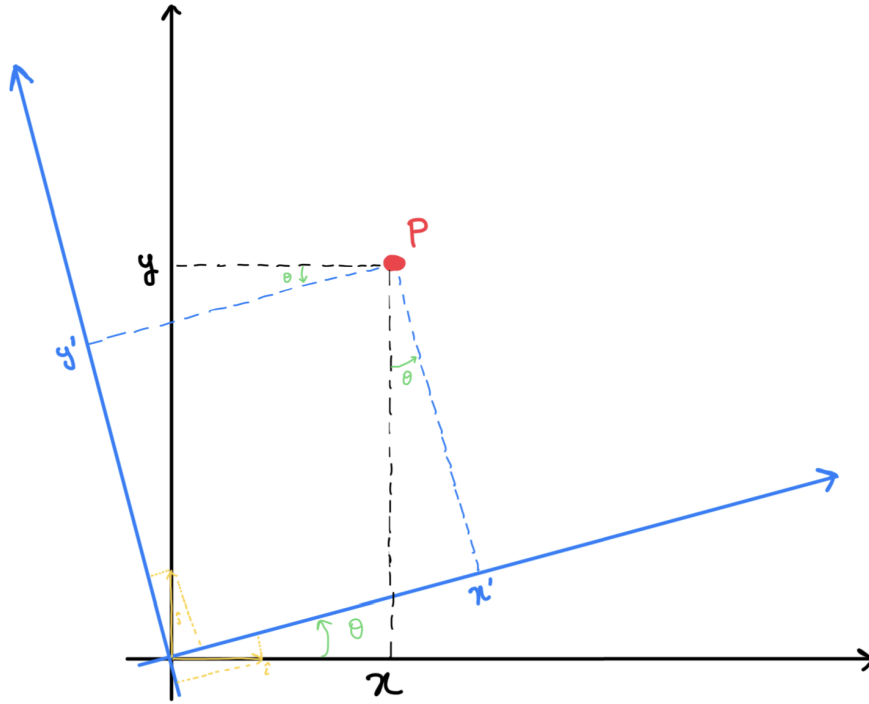


Figure 9: Given a point P in coordinate system (x, y) , how do you convert it to another coordinate system (x', y') which is tilted by an angle of θ counter clockwise?

Remember that this is a classic case of Linear Algebra. To Linearly Transform a point (x, y) to (x', y') , to first observe where the basis vectors of the original system go to in the new system (Marked in yellow in the figure).

$$(1, 0) \longrightarrow (\cos \theta, -\sin \theta)$$

$$(0, 1) \longrightarrow (\sin \theta, \cos \theta)$$

Therefore, a point (x, y) goes to $(x, 0) + (0, y) = x(1, 0) + y(0, 1)$

$$\implies (x, y) \longrightarrow x[\cos \theta, -\sin \theta] + y[\sin \theta, \cos \theta]$$

$$\implies (x, y) \longrightarrow [x \cos \theta + y \sin \theta, -x \sin \theta + y \cos \theta] = (x', y')$$

We can write this in the familiar Matrix form as

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

An easy way to remember this is as follows: The first basis vector $(1, 0)$ goes to the first column $\begin{bmatrix} \cos \theta \\ -\sin \theta \end{bmatrix}$ and the second basis vector $(0, 1)$ goes to the second column $\begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix}$.

6.2 Coordinates of Time

The standard approach to do this in Special Relativity is pretty straightforward. We imagine putting clocks at every point in space. When an event happens, the spatial coordinate at that point tell us “Where the event happened?” and the clock at that point tell us “When the event happened?”.

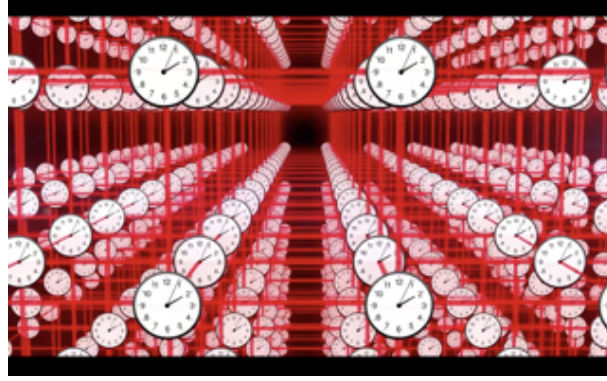


Figure 10: Clocks sitting at every single point in space.

6.2.1 How do we synchronize these clocks?

There is a subtle detail when we ask ourselves – “How do we synchronize the clocks in the first place?”. Our first guess would be to gather all the clocks at one place, synchronize them together and then move the clocks to their respective positions in space. Galileo would have been happy with this procedure. However we now know that clocks in motion tick off time differently. And hence, during the travel of each clock to its destined location, they all tick off time at different rates and this leads to asynchronism throughout.

The correct approach is as follows. We place the clocks (in off state) at their destined location. Then the clock at the origin sends out a bright flash radially outwards while starting its own clock. When the other clocks receive this flash, they start their clocks, but not from zero. They will account for the light travel time to them since they know their distance from the origin.

With this, we have laid out the requisites to create a coordinate system for space and time – **‘Spacetime Coordinates of an event’**. Before this, when we had to deal with position and time of an event, we had to undergo post processing. We had to account for the light travel time from where the event takes place till us to distinguish know when the event happened vs when we saw it happen. The benefit of having space-time coordinates is that we don’t need to do such a post processing. We just have to note down the space and time coordinates at the location and at the instant where the event happened.

(x, y, z, t) space-time coordinate of an event in 3D

(x, t) space-time coordinate of an event in 1D

Another benefit we derive from this is the simplicity when it comes to answering the 2nd question of the crux – “Where and When do someone else say an event happened?”. For this, we just need to look at the space and time coordinates of the event using the other person’s coordinate system. It might seem that we are neglecting the effects of Special Relativity with this methodology. But you have to trust me, the following example does hold true everytime.

setup: Two friends – Alice and Bob, are moving with respect to one another. They each have their own space-time coordinate systems. A distant firecracker goes off. For Alice, the position and time of the firecracker going off in *her* perspective is given by looking at *her* space-time coordinate readings at the location of the fire-cracker going off. More interestingly, Alice could look at the readings on Bob’s space-time coordinates at the location where the firecracker goes off and tell us exactly where and when Bob would say the event took place from *his* perspective. **Ultimately, Alice and Bob would read the exact same space-time coordinates for an event as long as they are both reading off the same coordinate system (either both reading Alice’s or both reading Bob’s).**

6.3 Asynchronous Nature of Clocks in Motion

Now we shall see how clocks which are synchronized from one frame of reference isn’t in sync from another frame if they are moving with respect to each other.

setup : We have two sets of coordinate system – one for train and the other for a platform moving relative to each other. Alice and Bob are on the moving train. Alice at the origin of the train coordinate system while Bob at x' of the train coordinate system. They are going to sync their clocks using the method we mentioned earlier. Bob initially sets his stopped clock at x'/c . Alice will send out a light beam and when Bob receives this beam of light, he starts his clock. From the train perspective, all is good and the clocks are in sync.

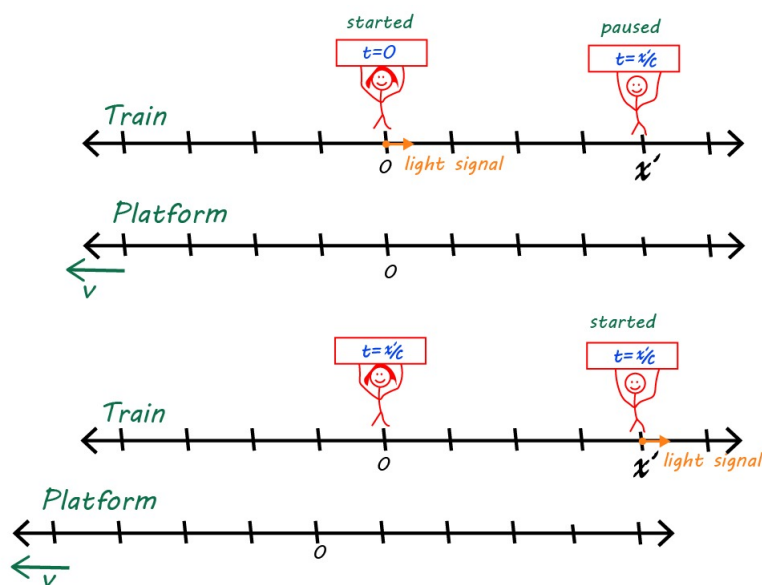


Figure 11: The clock syncing procedure from the train perspective is a success.

They can do the same for all clocks on the train and from the train perspective the synchronization procedure was successful. For simplicity, we shall only compare the clocks at the origin and Bob's location. Rest will follow the same pattern – either all in sync, or the amount of asynchronism will be linear with distance. (The intuition behind it being linear is pretty straightforward. If it weren't linear, then placing clocks at different intervals would lead to different results. The only graph which retains exactly the same shape and size when zoomed in or out is linear).

Do the observers on the platform also agree that synchronization was successful? For it to be successful, Alice and Bob's clocks should show the same time when the light signal passes him. They look at what time has elapsed on Alice's clock when the light signal passes Bob and compare it to what Bob's clock was set to start. The difference of these two times will give us the degree of asynchronism – $\Delta\tau$. If $\Delta\tau = 0$, that would imply that clocks are indeed simultaneous from both frames.

Flaws in the above synchronization procedure according to platform observers:-

1. The first fault they claim is the distance between Alice and Bob. Due to their motion, the distance between Alice and Bob undergo *length contraction* from $x' \rightarrow x'/\gamma$.
2. Since Bob is moving away from the light and the speed of light is constant, the light will take longer to reach Bob. Let Δt be the time taken by the light to reach Bob from the Platform perspective.

$$\Delta t = \frac{\frac{x'}{\gamma} + v\Delta t}{c}$$

$$\implies \Delta t = \frac{x'}{\gamma(c - v)}$$

3. The last fault that the platform observers claim is *Time Dilation*. Δt is the time that the platform observers say has elapsed from their perspective. However, this wouldn't be the time that reads on Alice's clock since she is in motion (Her clock ticks off time more slowly compared to ours). $\Delta t'$ be the time elapsed on Alice's clock when the light signal reaches Bob (as seen from someone on the platform). This should be what Bob's clock should have been set to if their clocks had to be synchronized.

$$\Delta t' = \frac{\Delta t}{\gamma}$$

$$\Delta t' = \frac{x'}{\gamma^2(c - v)}$$

$$\Delta t' = \frac{x'(c^2 - v^2)}{c^2(c - v)}$$

$$\Delta t' = \frac{x'(c + v)}{c^2} = \frac{x'}{c} \left(1 + \frac{v}{c}\right)$$

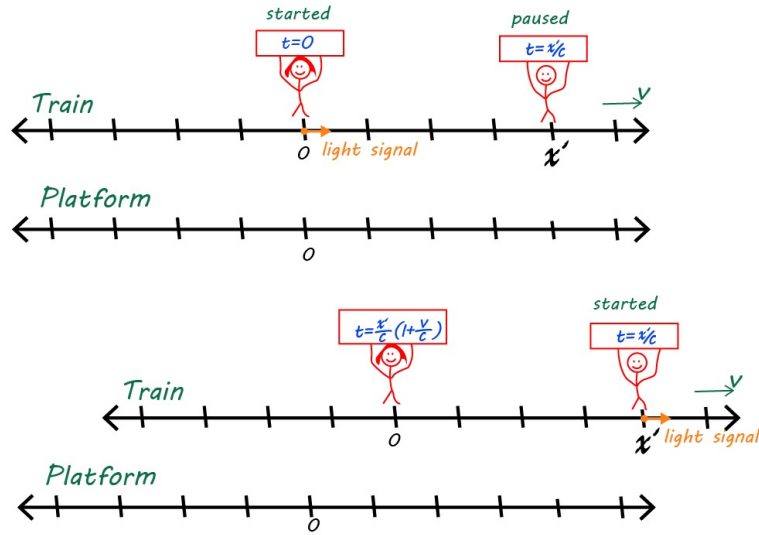


Figure 12: The clock synchronization procedure is a failure from platform perspective. Notice how the time on Alice's clock is different from the platform perspective.

From the platform perspective, the amount by which the time on Bob's clock differs from that on Alice's clock is given by the difference between time elapsed on Alice's Clock when the light signal passes Bob and the time set on Bob's clock originally

$$\Delta\tau = \frac{x'}{c} \left(1 + \frac{v}{c}\right) - \frac{x'}{c}$$

$$\boxed{\Delta\tau = \frac{x'v}{c^2}}$$

This is the key idea – **Clocks which are leading in motion are behind in time and those which are lagging in motion are ahead in time.** Synchronized clocks that are separated by a factor of x (measured in its own coordinate system) differ by xv/c^2 to another observer in relative motion.

Let's remind ourselves that we had discussed this lack of simultaneity earlier too (with a train). However, there we arrived at a different result. Which was

$$\Delta t = \frac{2Lv}{c^2 - v^2}$$

I now argue that the above result and the result we got just now are identical. How?

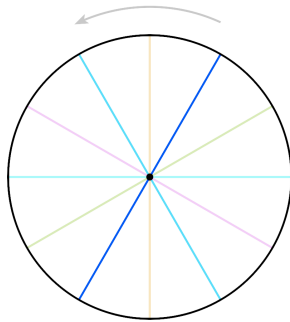
First of all, $2L$ is the length of the train as observed from the ground perspective meanwhile x' is the true length of the train. Hence, $2L = x'/\gamma = x'\sqrt{1 - v^2/c^2}$. Substituting that for Δt , we get

$$\begin{aligned}
\Delta t &= \frac{2Lv}{c^2 - v^2} \\
&= \frac{vx' \sqrt{1 - v^2/c^2}}{c^2 - v^2} \\
&= \frac{vx'}{c^2} \frac{1}{\sqrt{1 - v^2/c^2}}
\end{aligned}$$

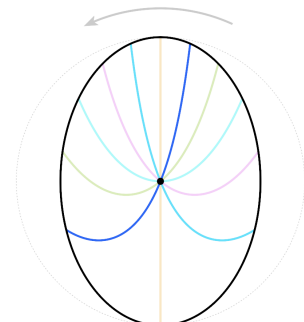
That was so close, we have roughly the same equation but with a small addition of γ factor. This were I got stuck for days.. I spent hours wondering why the discrepancy.. I went as far as to thinking like an engineer “maybe they used $v \ll c \implies \gamma \approx 1$ ”. But I knew it in my bones that it was not the case, it could never be.. And finally, after a few days of pondering I had my breakthrough understanding... Δt and $\Delta\tau$ are NOT the same! $\Delta\tau$ is the time difference read on the clocks of the moving frame between two locations meanwhile Δt is the time difference measured on ground clocks between two events that people on the ground say were supposed to happen at the same time. Would they be same? absolutely no.. why? Time dilation baby! The time difference measured on the ground Δt would be γ times the time difference that is read on the moving clocks $\Delta\tau$.

Asynchronous Clocks Example - Relativistic cycle wheel A wheel of a cycle moving incredibly fast deforms according to special relativity. One may only expect the circular wheel to look more oval shape due to Lorentz Contraction. However, in reality even the spokes seem to bend too like the picture below. How would you explain this?

$v = 0.00c$



$v = 0.72c$



Answer: This happens because of the asynchronous nature of time in motion. Taking the vertical spoke to be the reference (assuming wheel turns counter clockwise) spokes to the right of it are ahead in time (in the future of what it is supposed to be) and hence are bent closer to the center. Similarly, spokes to the left and behind in time and hence are also bent closer to the center.

6.3.1 Relativity of Simultaneity as a consequence of asynchronous clocks in motion

It is crucial to understand that lack of simultaneity of events with respect to a moving frame is a direct result of the asynchronous nature of clocks. An event that will happen at exactly 12:00 noon inside a moving train still happens at exactly 12:00 noon of the train's frame. However the time 12:00 noon is not absolute inside the train when viewed from outside. When the middle of the moving train is at 12:00 noon, the front may be at 11:55am and the rear may be at 12:05pm. Hence, the temporal order of events may change due to the asynchronous nature of clocks. As we shall see later, it is important to note that causality doesn't change. If event- A is the cause of event- B , B will always happen after A no matter what frame we choose. An example of this from an earlier discussion of Alex and Bob inside moving train waiting to receive light signals is as follows. From one frame, Alex and Bob may receive the light simultaneously and from another frame, Alex may get the light signal before Bob and in yet another frame, Bob may receive the light signal before Alex. But there doesn't exist any inertial frame where either one of them receives the light signal before it was sent out from the center of the train.

7 Lorentz Transformation

Now we are finally going to set up a dictionary to transform the space-time coordinates of one set of observers – Team Platform (x, t) , to the space-time coordinates of another set of observers – Team Train (x', t') . For simplicity, assume that when the origin of the frames passes each other, they set their clocks to zero. That is $(x = 0, t = 0)$ coincides with $(x' = 0, t' = 0)$.

7.1 Relating Spatial Coordinates

We focus on arbitrary point x' on the moving frame and ask ourselves – “Where is that point located on the stationary frame's coordinate system at a time t ?”

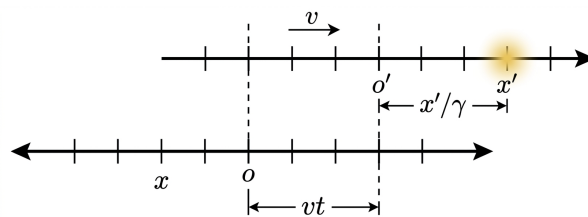


Figure 13: Measuring the moving length x' from the stationary frame gives us $\frac{x'}{\gamma}$

From the platform perspective, that distance is simply the sum of the distance moved by the moving coordinate system and the moving length x' as measured from the stationary frame. That is,

$$x = \frac{x'}{\gamma} + vt$$

Simplifying, I can write

$$\boxed{x' = \gamma(x - vt)}$$

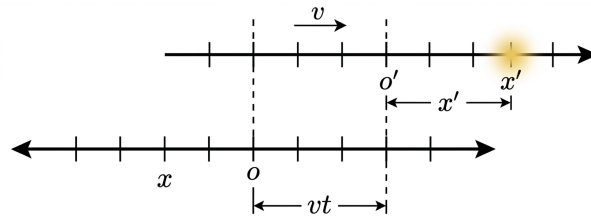


Figure 14: Remember that Newton for whom length contraction doesn't exist would arrive at $x = x' + vt$ and hence $x' = x - vt$

7.2 Relating Time Coordinates

There's no meaning when it comes to relating 'The time of the platform frame' to 'The time of the train frame' and vice versa. This is because all clocks on one frame are out of sync relative to observers on another frame. That is,

1. People on the platform say that clocks on the train are out of sync and
2. People on the train say that clocks on the platform are out of sync.

Hence, according to the stationary observers, the time on a clock on the moving frame depends on the position of the clock on that frame.

Let's look at what the people on the platform would say is the time on the moving train at a position x' when the clocks on the platform reads t . Note that from the platform perspective all clocks are in sync. Hence, there is no need to write t_{origin} .

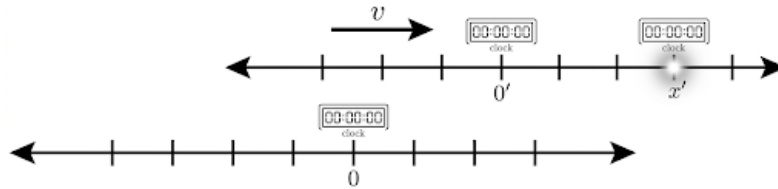


Figure 15: When the origin of the frames coincide, each send out a light signal that synchronizes each frame's clocks. We are observing this from the platform perspective. Hence, all clocks on the platform are in sync meanwhile those on the train are out of sync.

First of all, the clocks at the origin of both frames agree at $t = 0$. That is,

$$t = 0 \implies t'_{\text{origin}} = 0$$

First we only compare the time on the origins of the two coordinate systems so that we only have to deal with time dilation. From the platform perspective, clocks on the moving train are ticking off time more slowly and hence

$$t = \gamma \times t'_{\text{origin}}$$

Let us put that aside and remember how we related the time difference between asynchronous clocks. Firstly, leading clocks are behind in time and therefore, the clock at x' will show a smaller time than that at origin. The magnitude of the difference was $\frac{x'v}{c^2}$ where x' is measured in its own coordinate system.

Therefore,

$$t'_{x'} = t'_{\text{origin}} - \frac{x'v}{c^2}$$

$$\Rightarrow t'_{\text{origin}} = t'_{x'} + \frac{x'v}{c^2}$$

Substituting this in the earlier equation, we get...

$$t = \gamma \left(t'_{x'} + \frac{x'v}{c^2} \right)$$

This is what we've been looking for. However if I have to write t' in terms of t and x . Then, the logic is pretty interesting. Team train would say that they are stationary and it is the platform folks that are moving backwards. Hence, replace all primed variables with unprimed and velocity becomes $-v$ instead of $+v$.

$$t' = \gamma \left(t - \frac{xv}{c^2} \right)$$

To avoid confusion take the 'primed reference frame' to be the one that is moving towards the positive axes of the other coordinate frame. Then v would just be the velocity given in the question. Else, take the negative of it.

What we have now is a dictionary – A mindless approach, given when and where an event takes place for one set of observers, we just plug in the values and obtain the same for the other set of observers. All effects of special relativity that we have discussed so far have been included in these formulae.

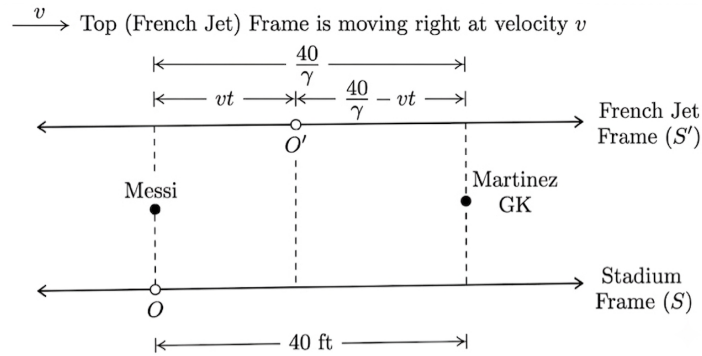
7.3 Practice Problems on Lorentz Transformations

Lorentz Transformation Example - FIFA World Cup 2022

Argentina wins the 2022 FIFA World Cup final against France. To celebrate, Lionel Messi stands at the center spot and fires a celebratory football towards his goalkeeper – Emiliano Martínez, with a speed of $\frac{4}{5}c$. Since Martínez is currently near the goal line 40 foot away, the ball hits him exactly 50 nanoseconds later. The French team, having just lost, are quickly leaving Qatar in their high-speed jet at $v = \frac{3}{5}c$ in the same direction as that of the ball. They started from the exact spot and instant messi kicked the ball. Where and when do the French players say Martínez caught the ball?

Answer: I shall do this question in two ways. First by thinking it through – The method of first principles, and the second by directly applying Lorentz transformation to blindly get the answer I’m looking for. Given the velocity $v = \frac{3}{5}c$, I calculate $\gamma = \frac{5}{4}$

1. Going by the method of first principles, we might be tempted to apply time dilation directly to the 50 nanoseconds that we see. However remember always that you shouldn’t apply time dilation directly when dealing with events that happen at different locations. Here’s the correct procedure...



From the French people’s perspective, the stadium is moving backwards at $\frac{3}{5}c$ and hence the clock at Martínez’s spot is ahead in time by an amount $\frac{vx}{c^2}$ with respect to the clock at Messi’s spot (origin of stadium frame). From our perspective, when the ball hits the GK, the time on the GK’s clock show 50 nanoseconds and at the same instant, time at the origin shows

$$t_o = t_{\text{GK}} - \frac{vx}{c^2}$$

$$t_o = 50 - \frac{\frac{3}{5} \cdot 40}{1^2} = 26 \text{ ns}$$

Here on we can directly apply the time dilation formula. We need to find the amount of time that elapses on our clock for which 26 nanoseconds passes on the origin clock of the stadium frame

$$t' = \gamma \times t_o$$

$$t' = \frac{5}{4} \times 26 = \frac{65}{2} \text{ ns}$$

We still sit on the French people’s frame and try to find the position of Martínez when the ball hits him. We first look at what the Lorentz contracted length between Messi and Martínez is. We get it to be $\frac{40}{\gamma} = 32$ foot. Out of this 32 foot, the stadium frame already moved $\frac{3}{5} \times \frac{65}{2} \text{ ns} = 19.5$ foot back. The remaining distance to Martínez from our origin on our coordinate system $\Rightarrow x' = 12.5$ foot.

2. Applying Lorentz transformation directly is more straightforward. We see two events here

Event 1 : Ball is kicked towards GK	Event 2 : GK catches the ball
Stadium : $(x = 0, t = 0)$	Stadium : $(x = 40 \text{ ft}, t = 50 \text{ ns})$
French : $(x' = 0, t' = 0)$	French : $(x' = ?, t' = 0?)$

The primed reference frame is moving towards the positive side of unprimed reference frame and hence $v = +\frac{3}{5}c$.

$$x' = \gamma(x - vt) = \frac{5}{4} \left(40 - \frac{3}{5} \times 50 \right) = \frac{50}{4} \text{ foot}$$

$$t' = \gamma \left(t - \frac{vx}{c^2} \right) = \frac{5}{4} \left(50 - \frac{\frac{3}{5} \cdot 40}{1^2} \right) = \frac{65}{2} \text{ ns}$$

As we can see, both methods yield the same answer. The second method is very straightforward meanwhile the first method is more fun.

Lorentz Transformation Example - Track Race

Imagine a sprinter is running a 3000 foot race along a straight track. The sprinter is wearing a watch that starts ticking from zero when the starting gun goes off. The track is equipped with a clock at the start line and another at the finish line, to record the elapsed time for the sprinter to cover the 3000 foot distance. Ignore unnecessary complexities like the travel time from the gun to the sprinter's ear and the fact that the sprinter is accelerating from rest. Assume that the sprinter has a constant speed $v = \frac{3}{5}c$ throughout. Use approximation that $c = 1 \text{ ft/ns}$ and answer the following questions.

- (i) How long will it take for the sprinter to finish the race according to the stadium clocks?

Answer: According to stadium clocks, the sprinter covers 3000 ft and is traveling at a speed of $\frac{3}{5}c$. Calculating time is then fairly straightforward

$$\Delta t_{\text{stadium}} = \frac{3000}{3/5} = 5000 \text{ ns}$$

- (ii) According to the sprinter, what is the distance between the start and finish lines?

Answer: From the sprinter's perspective, he is stationary and the track moves back at $\frac{3}{5}c$ and hence the distance between start and finish becomes lorentz contracted

$$\Delta x' = \frac{\Delta x}{\gamma} = \frac{3000}{5/4} = 2400 \text{ ft}$$

(iii) How long will it take for the sprinter to finish the race according to his own clock?

Answer: From the sprinter's perspective, the 2400 ft track (according to his perspective) is moving backwards at $\frac{3}{5}c$ and hence the time elapsed from his frame is given by

$$\Delta t_{\text{sprinter}} = \frac{2400}{3/5} = 4000 \text{ ns}$$

(iv) According to the sprinter, how much time elapses on all stadium clocks during the race?

Answer: From the sprinter's perspective, the clocks on the track are out of sync. However the time that elapses on each of the clocks would be the same. Due to the apparent motion of the clocks, time is ticking off slower by a factor of $\frac{5}{4}$. Hence, the sprinter would say that $\frac{4000}{5/4} = 3200$ nanoseconds would pass on both the clocks on the track.

(v) According to the sprinter, are the stadium clocks properly synchronized? If not, what is the reading on the clock at the finish line when the clock at the start line reads 0?

Answer: No! From the sprinter's perspective, the clocks on the track are not in sync. Since the track appears to move backwards, lagging clocks which in this case is the clock at the finish line will be ahead. By a factor of $\frac{vx}{c^2} = \frac{\frac{3}{5} \times 3000}{1^2} = 1800$. Hence, when the clock at the start line reads 0, the one at the end line shows 1800 ns.

(vi) According to the sprinter, what are the readings on the stadium clocks located at the start line and at the finish line at the end of the race? Do the sprinter and the stadium officials agree on the reading of the stadium's clock at the finish line when the race concludes?

Answer: At the start of the race, from the sprinter's perspective, the clock at the start line will read 0 ns. While that on the finish line will read 1800 ns. At the end of the race, the clock on the start line will read 3200 nanoseconds and the one in the finish line which is always ahead by 1800 ns will read 5000 nanoseconds. Hence, the sprinter and stadium officials will agree on the reading of the stadium's clock at the finish line when the race concludes.

Lorentz Transformation Example - Broomstick-1

Consider two broomsticks whose resting lengths are each 255 feet with the only difference being that one is red and the other is blue. From your perspective on Earth, the blue broomstick is stationary and the other red broomstick is rushing by in the positive x-direction with speed $v = \frac{8}{17}c$. Answer the following questions

(i) From your perspective, what is the length of the red broomstick? From the perspective the observer riding the red broomstick, how long is your blue broomstick?

Answer: From our perspective, we are at rest and the red broomstick is rushing past

us at $v = \frac{8}{17}c$ which corresponds to a γ of $\frac{17}{15}$. The same applies to observing the blue broomstick from the perspective of red broomstick. Hence, the lorentz contracted length becomes

$$L = \frac{255 \text{ ft}}{\gamma} = 225 \text{ ft}$$

(ii) Each broomstick is equipped with a light on each of its ends. Observers that are riding the red broomstick flash both of its lights simultaneously, from their perspective. How far apart do you say the flashes took place?

Answer: From our perspective, the flashing of lights doesn't take place simultaneously. If we assume the first light going off to be at the origin at $t = 0$, then what we want is the position of the second light going off from our perspective. General form of Lorentz transformation is $x' = \gamma(x - vt)$. However since the broomstick is moving forward from our perspective, that is the primed reference frame. We shall invert it and write

$$\begin{aligned} x &= \gamma(x' + vt') \\ \Rightarrow x &= \frac{17}{15} \times \left(255 + \frac{8}{17} \cdot 0 \right) = 289 \text{ ft} \end{aligned}$$

Another way of doing this problem is by measuring the time difference and seeing how much the broomstick moves in this interval.

$$\Delta t = \frac{xv}{c^2} = 255 \times \frac{8}{17} = 120 \text{ ns}$$

This is the time difference on the clocks in rear and front of the broomstick according to us. But actual time elapsed on our clocks is different because we claim that Red Broomstick's clocks only ticked off 120 ns because their clocks tick off time more slowly. Actual time elapsed between the lights going off would then be $120 \times \gamma = 136$ ns.

During this interval, the broomstick moves 64 ft and since the apparent length of the broomstick from our perspective is 225 ft, we say that the light flashes took places $64 + 225 = 289$ ft apart.

Appreciate how both answers are in perfect agreement !

(iii) You flash the lights on your blue broomstick simultaneously. According to those riding the red broomstick, how far apart did these flashes take place?

Answer: From Red Broomstick's perspective, it is at rest and it is the blue broomstick that is moving. Hence everything in the previous sub-question applies here too in perfect match. Thus, 289 ft apart.

(iv) As the team of observers riding along with the red broomstick pass by you, they grab your blue broomstick in a manner that's simultaneous from their perspective.

When they subsequently compare the length of the red and blue broomsticks, will they say that the blue one is longer than the red, shorter than red, or equal to red?

Answer: Shorter! because from Blue people's perspective the act of measuring was not simultaneous. First they grabbed the front of the blue broomstick, let it move for a while and then grabbed the rear.

(v) Going back to the original set up of the problem, with the blue broomstick at rest from your perspective and the red moving in the positive x-direction with speed $\frac{8}{17}c$. what speed must a third frame of reference have, from your perspective, in order that observers in this new frame claim that both broomsticks have the same length?

Answer: Our first instance would be to take a reference frame moving half as fast as the red broomstick. Thus $\frac{4}{17}c$. But this will lead to an error because remember how addition of velocities happen more complicatedly in special relativity. We want to find the velocity v for which we'd say that the frame is moving forward with v (since we are taken to be at rest) and that frame would say that the red broomstick is also moving at v . Recalling,

$$v_{2,1} = \frac{v_2 - v_1}{1 - \frac{v_1 v_2}{c^2}}$$

we apply

$$\begin{aligned} v &= \frac{\frac{8c}{17} - v}{1 - \frac{8v}{17c}} \\ \implies \frac{8}{17c}v^2 - 2v + \frac{8c}{17} &= 0 \\ \implies v &= 0.25c \end{aligned}$$

For a frame that moves at $0.25c$, both the blue broomstick and the red broomstick will appear to move at the same *speed*. And hence their corresponding γ would be same and so will be the lorentz contracted length.

Lorentz Transformation Example - Broomstick-2

Imagine a team of observers (Team Red) are riding a red broomstick, and zip by Earth at speed v . Assume the origins of Team Earth and Team Red cross at $t = t' = 0$. Imagine that a member of Team Red located at their origin has stolen a top secret file. So, you jump on your new super-broomstick, and chase after Team Red at speed V such that ($V > v$). You leave Earth at time $t = t_1$. (To distinguish the various γ factors, write $\gamma[w] = \frac{1}{\sqrt{1-w^2}}$ for any w expressed in units where $c = 1$).

(i) At what time t_2 according to Earth clocks do you catch up with the outlaw in Team Red?

Answer: From the Earth frame, for us to catch up to the outlaws, the distance

traveled by us in $(t_2 - t_1)$ time should be equal to the distance traveled by the outlaws in t_2 time

$$\begin{aligned} vt_2 &= V(t_2 - t_1) \\ \implies vt_2 &= Vt_2 - Vt_1 \\ \implies t_2 &= \frac{Vt_1}{V - v} \end{aligned}$$

Notice how relativistic effects don't come into picture here because we are measuring the events that take place according to our own clocks.

(ii) What is the time according to Team Red when you start chasing after them?

Answer: According to the Red Broomstick folks, earth clocks are ticking off time more slowly. Hence, by the time the clock at the origin of earth hits t_1 , a time of $\gamma[v]t_1$ would have elapsed from the frame of reference of Red Broomstick.

Doing the same by Lorentz transformation, (Team Red being the primed frame),

$$t' = \gamma[v] \left(t - \frac{vx}{c^2} \right)$$

From our frame, when we left earth, our clocks read $t = t_1$, and our $x = 0$.

$$\implies t' = \gamma[v]t_1$$

(iii) According to Team Red, how far away are you from the outlaw when you start chasing them?

Answer: According to Team-Red, they have traveled for a duration of $\gamma(v)t_1$ before we started to chase them. Hence from their perspective, they'd have covered a distance of $v_1 t_1 \gamma[v]$.

Doing the same by Lorentz transformation, (Team Red being the primed frame),

$$x' = \gamma[v](x - vt)$$

From our frame, when we left earth, our clocks read $t = t_1$, and our $x = 0$.

$$x' = \gamma[v]v t_1$$

(iv) According to Team Red, at what time do you catch up with the outlaw?

Answer: Let $\Delta t'$ be the time elapsed from th perspective of Team Red from the point you left earth till you catch up to them. Then

$$\begin{aligned} \left(\frac{V-v}{1-vV} \right) \Delta t' &= \gamma[v]v t_1 \\ \implies \Delta t' &= \frac{(\gamma[v]v t_1)(1-vV)}{V-v} \end{aligned}$$

Let t'_2 be the time according to Team Red when we catch up to the outlaw. Then $t'_2 = t'_1 + \Delta t'$,

$$\begin{aligned} t'_2 &= \gamma(v)t_1 + \frac{(\gamma[v]v t_1)(1-vV)}{V-v} \\ \implies t'_2 &= \gamma[v]t_1 \left(1 + \frac{v(1-vV)}{V-v} \right) \\ \implies t'_2 &= \frac{t_1}{\sqrt{1-v^2}} \left(\frac{V(1-v^2)}{V-v} \right) \\ \implies t'_2 &= \frac{Vt_1}{\gamma[v](V-v)} \end{aligned}$$

Doing by Lorentz transformation, we have an event (we catch upto the outlaw) in the earth frame that happens at $t = t_2$ at the location $x = x_2 = vt_2$

$$\begin{aligned} t'_2 &= \gamma \left(t_2 - \frac{vx_2}{c^2} \right) \\ \implies t'_2 &= \gamma[v](t_2 - v^2 t_2) = \frac{t_2(1-v^2)}{\sqrt{1-v^2}} \\ \implies t'_2 &= \frac{t_2}{\gamma[v]} = \frac{Vt_1}{\gamma[v](V-v)} \end{aligned}$$

(v) According to your watch (which read t_1 when you left Earth), what time is it when you catch up with the outlaw?

Answer: At the beginning of the journey ($t = t_1$), our clock was in sync with earth clock at origin. Let $\Delta t''$ be the time that elapses on our clock from the moment we leave earth till we catch up to the outlaw. At the beginning of the journey ($t = t_1$), our clocks were in sync with earth clocks. The earth observers would then look at our clock and claim that we are in motion and hence our clock should tick off time more slowly. That is, $\Delta t'' = \Delta t / \gamma[V]$. Since $\Delta t = \frac{vt_1}{V-v}$,

$$\Delta t'' = \frac{vt_1}{\gamma[V](V-v)}$$

Now, $t''_2 = t_1 + \Delta t''$

$$\begin{aligned}\Rightarrow t_2'' &= t_1 + \frac{vt_1}{\gamma[V](V-v)} \\ \Rightarrow t_2'' &= t_1 \left(1 + \frac{v}{\gamma[V](V-v)} \right)\end{aligned}$$

In the Earth's frame, Δt is calculated by taking the reading on the clock at the origin and subtracting it from the time read on the Earth frame clock at the location and instant where we pass the outlaw. While these two clocks are perfectly synchronized in the Earth's frame, they are not in sync from our frame.

In this sub-question, the reason why we can't do "earth is in motion and hence its clocks should tick off time more slowly and therefore by the time Δt elapses on earth clocks, $\gamma[V]\Delta t$ would elapse on our clocks", is because from our perspective it is the earth-frame's clock at the location of us catching the outlaw that reads a time $t = t_2$, not the clock at the origin. Hence from our frame, the time elapsed on the origin clock of earth's frame is not Δt .

Lorentz Transformation Example - Train Assassination

Consider two frames of reference, Team Train and Team Platform. From the perspective of Team Platform, Team Train is moving east at speed $v = \frac{12}{13}c$. Each team properly synchronizes their clocks, and sets things up so that when the clocks at the origin of each frame of reference pass one another, they both read 4 PM. Let's now imagine that a devastating crime is committed: President of Backward-land, riding along in Team Train, is assassinated. (President of Backward-land is facing backwards in the train, i.e. he is located at the front of the train and is facing the rear.) An observer in Team Platform, who happened to be situated where the assassination took place, notes that the clock in Team Train (at the location of the crime) reads 2 PM when the crime was committed. This same observer notes that the clock in Team Platform (at that same location) reads 4 PM when the crime was committed. Express all answers in the units of hours for time and light-hours for distance.

(i) According to those in Team Train, when the President was assassinated what was the reading on the clock in their frame of reference that is located where the crime was committed?

Answer: 2:00 PM

(ii) According to those in Team Train, what was the reading on the clock at the origin of Team Train when the crime was committed?

Answer: According to Team Train, all clocks on the train are in sync. Hence even the clock at the origin of train frame would read 2:00 PM

(iii) According to those on Team Platform, what was the reading on the clock at the origin of Team Platform when the crime was committed?

Answer: 4:00 PM

(iv) According to those on Team Platform, what was the reading on the clock at the origin of Team Train when the crime was committed?

Answer: 4:00 PM

(v) According to those on Team Train, what is the reading on the clock at the origin of Team Platform when the crime is committed?

Answer: Given that the origin clocks on Team Train and Team Platform were synchronized at 4:00 PM, we need to find out the reading on the origin of Team Platform when the origin of Team Train reads 2:00 PM. Since that was 2 hours ago from the Team Train's perspective and less time would have elapsed for the Platform clocks from Team Train's perspective. The clock at the origin of team platform was only $\frac{2 \text{ hrs}}{\gamma} = 0.768 \text{ hours} = 46 \text{ min}$ behind 4:00 PM. That is, the clock on origin of team platform would read 3:14 PM.

(vi) Determine where, according to Team Train, the origin of Team Platform was located when the crime was committed. Express your answer in light-hours from the origin of Team Train.

Answer: From Team Train's perspective, the origin of team train has not yet reached team platform's origin because that will only happen when all clocks on team train reads 4:00 PM from Team Train's perspective. Since now the time at the origin of the train is 2:00 PM, the origin of the train needs to travel 2 hours more to reach the platform.

The distance traveled by the train in this interval is $2 \times 12/13 = 24/13$ light-hours. Thus, right now, the origin of team platform is located at $\frac{+24}{13}$ light-hours away from the origin of team train.

(vii) Determine where the assassination took place according to those on Team Train. Express your answer in light hours from the origin of Team Train.

Answer: If from the Platform perspective, there is a 2 hour time difference between the origin of the train and location of assassination, I can use $\Delta t = xv/c^2$ to see that both points are separated by a distance of $13/6$ light-hours measured using the train's spatial coordinates.

(viii) Similarly, determine where the assassination took place according to those on Team Platform.

Answer: From the Team Platform's perspective, the origin of Team Train and Team Platform coincides when the assassination takes place and also the length of the train undergoes Lorentz contraction. Hence, from the platform perspective, the assassination takes place at $\frac{13/6}{\gamma} = \frac{5}{6}$ light hours from the Platform Origin. The sign will be positive because the location where assassination took place is behind in time compared to the origin.

(ix) When the observer on Team Platform situated at the location of the assassination sees the crime committed, he immediately sends a light signal to the observer at the origin of Team Platform, alerting him to what happened. At what time does the observer at the origin of Team Platform receive the message?

Answer: Light takes $5/6$ hours to reach the platform origin. Hence he receives the signal at 4:50 PM

(x) The observer on Team Train who happened to be located where the crime was committed also immediately sends out a light signal alerting his colleagues of the crime. At what time does the observer at the origin of Team Train receive the message?

Answer: Light takes $13/6$ hours to reach the origin of Team Train. Hence, they get alerted $2\frac{1}{6}$ hours after 2:00 PM. Hence, at 4:10 PM.

(xi) When the observer at the origin of Team Platform receives the message from his colleague that was located at the crime scene, he immediately tells the member of Team Train who happens to be passing him at that moment. What does this observer in Team Train say her clock reads at that moment?

Answer: First we need to find the point on the train passing through the platform's origin when the clock at the platform origin reads 4:50 PM. The train has traveled for $\frac{5}{6}$ hours since the origin of Team Train coincided with the origin of Team Platform.

Thus, the origin of the train is at $\frac{5}{6} \times \frac{12}{13} = \frac{10}{13}$ light-hours. Does that mean the point directly opposite to Team-Platform will read -10/13 light-hours on the Train frame?. Absolutely not! This is because lengths on team train are shortened from Team Platform's perspective. Hence a distance of $\frac{-10}{13}$ from the train's origin would read as $\gamma \times \frac{-10}{13} = -2$ light-hours. At this instant, the clock at the origin of Team Train would read $\frac{5/6}{\gamma} = \frac{25}{78}$ hours elapsed since 4:00 PM ie 4:19PM from our frame. Also, the clock at the origin of Team Train is out of sync with the clock at -2 light-hours on the Train-frame by a factor

$$\Delta t = \frac{vx}{c^2} = \frac{12}{13} \times -2 = \frac{-24}{13} \text{ hours}$$

The negative sign indicates that the clock at -2 light-hours is 'ahead' in time with respect to the origin clock by 1 hour 51 minutes. Hence, the time at -2 light-hours will read 6:10 PM.

To avoid all this excessive thought processes, we could have just directly applied Lorentz Transformation. $\Delta t'_{(-2)}$ be the time that elapses on the point -2 light-hours on the train frame when $5/6$ hours have passed on the origin of Platform frame.

$$\Delta t'_{(-2)} = \gamma \left(\Delta t - \frac{vx}{c^2} \right)$$

since we are dealing with the origin of platform frame, $x = 0$ and $\Delta t = 5/6$. Hence,

$$\Delta t'_{(-2)} = \frac{13}{5} \times \frac{5}{6} = \frac{13}{6} \text{ hours} = 2\text{hour } 10\text{min}$$

$$4:00 \text{ PM} + 2 \text{ hour } 10 \text{ min} = 6:10 \text{ PM}$$

(xii) When the observer at the origin of Team Train receives the signal in part (x), how far away will he/she claim he/she is from the observer at the origin of Team Platform?

Answer: Observer at the origin of Team train receives the light signal at 4:10 PM. That is, $\frac{1}{6}$ hours after the origins coincide. According to Team Train's perspective, the Platform would have moved $\frac{1}{6} \times \frac{12}{13} = \frac{2}{13}$ light hours. Hence, the platform origin is $2/13$ light hours away from the origin of Team Train measured using the train's coordinate.

(xiii) If the observer at the origin of Team Train immediately forwards (via another light signal) the light signal she received (part (x)) to the observer at the origin of Team Platform, at what time will that observer say the signal arrived?

Answer: Although from the train perspective, the light signal was sent out from the Train's origin at 4:10 PM, the platform observers would say that this event took place at 4:26 PM. From the Train Perspective, the platform origin is $2/13$ light-hours away from the observer when he sends out the light signal. The Platform observer would claim that the distance is actually $\gamma \times 2/13 = \frac{2}{5}$ light-hours away. The speed of the light signal from the platform perspective is c itself and hence light takes $\frac{2}{5}$ hours = 24 mins to reach him. Therefore, the platform observer reads 4:50 PM on his clock when the light signal reaches him.

(xiv) Consider the clock in Team Train which, according to Team Platform, read 2 PM when the crime was committed. According to Team Platform, what time will it be on Team Platform's clocks when this train clock strikes 4 PM?

Answer: According to Team Platform, when the crime was committed on Team Train, it read 4:00 PM on the platform clocks. For it to read 4:00 PM on that clock on the train, 2 hours have to elapse on the train clocks. It takes $\gamma \times 2$ hours on the platform clocks for 2 hours to elapse on the train clocks. Hence, Team Platform's clocks will read 9:12 PM.

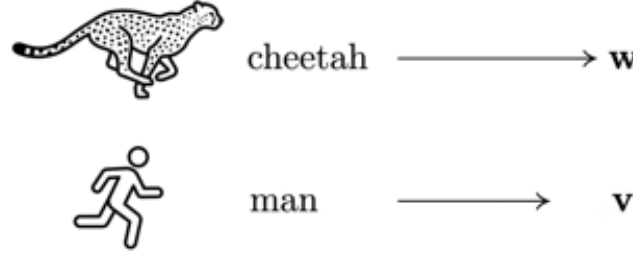
8 Combining Velocities

8.1 Combining Velocities in 1-D

We saw earlier how the Galilean method of addition of velocities in 1-D will lead to problems when involving speeds close to the speed of light or speed of light itself. Which is why we had to come up with the new formula for the addition/subtraction of velocities in Special Relativity

$$v_{1,2} = v_1 - v_2 \quad \longrightarrow \quad v_{1,2} = \frac{v_1 - v_2}{1 - \frac{v_1 v_2}{c^2}}$$

$$v_{1,2} = v_1 + v_2 \quad \longrightarrow \quad v_{1,2} = \frac{v_1 + v_2}{1 + \frac{v_1 v_2}{c^2}}$$



Our job now is to derive these results using Lorentz Transformation. Imagine a cheetah and a man racing. The cheetah sprinting at w and the man running at v . We are on the ground observing this.

From our frame, in time t , the cheetah covers a distance of wt and the man covers a distance of vt (Assuming they both started at $(0,0)$).

The Cheetah's spacetime coordinates from the ground frame is therefore (t, wt) and the man's is (t, vt) .

Now, I want the space-time coordinates of the Cheetah from the man's perspective. Applying Lorentz transformation taking the man as primed frame and us as unprimed, we get

$$\begin{array}{l|l} t' = \gamma \left(t - \frac{vx}{c^2} \right) & x' = \gamma(x - vt) \\ \Rightarrow t' = \frac{1}{\sqrt{1 - v^2/c^2}} \left(t - \frac{vwt}{c^2} \right) & \Rightarrow x' = \frac{1}{\sqrt{1 - v^2/c^2}} (wt - vt) \end{array}$$

From the Man's perspective, the cheetah will be at position x' at a time t' . Also, the speed of the cheetah was constant throughout and therefore the speed of the cheetah from the man's perspective is

$$v_{\text{rel}} = \frac{x'}{t'} = \frac{wt - vt}{t - \frac{vwt}{c^2}} = \frac{w - v}{1 - \frac{wv}{c^2}}$$

Hence, we have derived the relativistic form of addition and subtraction of velocities from Lorentz Transformation. For the case of Cheetah and Man running in opposite directions, replace $+v$ with $-v$ and you'll get the correct result.

8.2 Combining Velocities in 3-D

Now we want the general case. When the object are not moving directly towards each other or away from each other but in 3D space. This might seem like just another academic

exercise to show that we can extend our knowledge of 1-D to two and three dimensions. However, it is much more than that. We eventually see why this is important for deriving mass-energy equivalence.

Let's say I am on a train heading towards the positive x -direction with speed v . There is an object moving in space with velocity (w_x, w_y, w_z) . Our goal is to find the relative velocity of this object from our frame on the train. Once again, we have our Platform perspective and also, all the origins coincided at $t = 0$. We are now going to set up the space-time coordinates (t, x, y, z) of the object at time t from the frame of reference of the Platform.

From the Platform perspective, the object is at $(t, w_x t, w_y t, w_z t)$. We can use Lorentz Transformation to convert the following to the Train's Coordinate system (t', x', y', z') .

One might assume that γ , can be split into 3 directions as $\gamma[v_x]$, $\gamma[v_y]$ and $\gamma[v_z]$. But this is wrong because even if the train had velocities in 3 directions (v_x, v_y, v_z) , there is only one Lorentz gamma factor and it equals $\frac{1}{\sqrt{1-(v_x^2+v_y^2+v_z^2)/c^2}}$. Here, $\gamma[v]$ refers to $\frac{1}{\sqrt{1-v^2/c^2}}$. Hence, here we get...

$$\begin{aligned}x' &= \gamma[v](x - v_x t) = \gamma(w_x t - vt) \\y' &= w_y t \\z' &= w_z t \\t' &= \gamma[v] \left(t - \frac{vx}{c^2} \right) = \gamma \left(t - \frac{vw_x t}{c^2} \right)\end{aligned}$$

From the Train perspective, the object is at $(\gamma(t - \frac{vw_x t}{c^2}), \gamma(w_x t - vt), w_y t, w_z t)$.

From this, we can evaluate w'_x , w'_y and w'_z just as we did before :

$$\begin{aligned}w'_x &= \frac{x'}{t'} = \frac{w_x - v}{1 - \frac{vw_x}{c^2}} \\w'_y &= \frac{y'}{t'} = \frac{w_y}{\gamma(1 - \frac{vw_x}{c^2})} \\w'_z &= \frac{z'}{t'} = \frac{w_z}{\gamma(1 - \frac{vw_x}{c^2})}\end{aligned}$$

It is interesting to note that the relative velocity of the object in the y and z direction **does change** even though the train is only moving in the x -direction. This happens because of the difference in the rate at which time elapses for the the Train and the Platform. Eventhough distance traveled by the object in the transverse directions is the same for Train observers as that of Platform observers, time elapsed is different and hence relative speed also differs.

9 Space-time Diagrams

Space time diagrams are pictorial representation of the flow of time and position of events or motion of objects through space. It contains one time axis(usually the z -axis and the remaining spatial axes corresponding to every point in space for every point in time.

Space-time Diagrams are hence nothing but a collection of ‘now slices’ arranged one over the other. Each slice has information about all events having at that one instant in all of space.

Imagine drawing the space-time diagram of flash of light going off at $(0,0)$ and spreading circularly in all directions. We choose our units carefully so that distances in the xy -plane are measured in ct while time in z -axis is measured in t . This would imply that the 3-D (2 space + 1 time) space-time diagram of a flash of light going off would look like an infinite upward cone inclined to the z -axis at 45° .

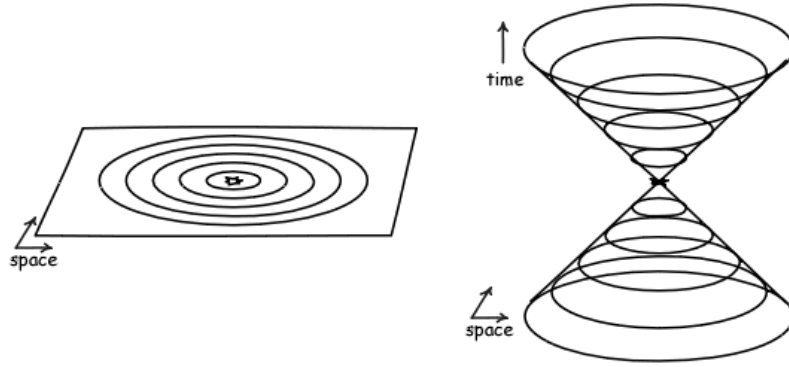


Figure 16: This particular space-time diagram is also called Light-Cone.

9.1 Space-time Diagrams of two observers in motion

Now we will look at the space-time diagrams of two observers who are in relative motion. We had discussed before under 9 how to transform from one set of coordinates to another. There we dealt with transforming $(x, y) \rightarrow (x', y')$. Interestingly, transformation of $(x, t) \rightarrow (x', t')$ follows a similar pattern conceptually. Therefore Lorentz transformation can be thought of as changing the coordinate axes in a space-time diagram.

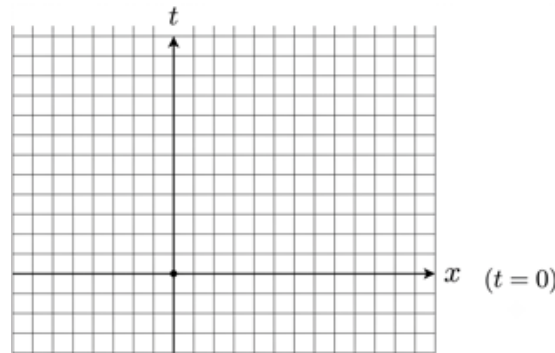


Figure 17: The space-time diagram for a single observer.

Note that x -axis here refers to the line for which $t = 0$. Therefore, to find x' -axis, we just need to find the equation of the line for which $t' = 0$. From Lorentz Transformation (I take $c = 1$), we know...

$$t' = \gamma(t - vx)$$

$$\implies t' = 0 \text{ for } t = vx$$

Equation of x' -axis is therefore $t = vx$. A straight line with slope v ($v < 1$, because $c = 1$). Similarly, t' -axis is the line where $x' = 0$.

$$x' = \gamma(x - vt)$$

$$\implies x' = 0 \text{ for } x = vt$$

Equation of t' -axis is therefore $t = \frac{1}{v}x$. A straight line with slope $1/v$.

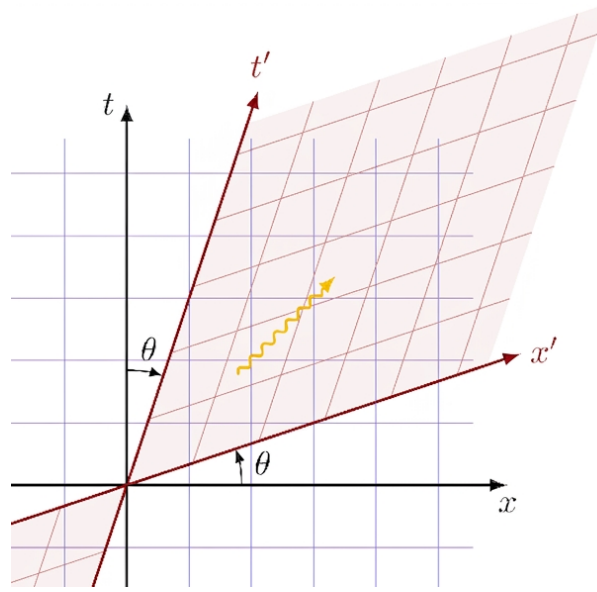


Figure 18: Space-time diagram for two observers in motion. Here, $\tan \theta = v$

Thus, the same event (a point in space-time cartesian plane) can be expressed in (x, t) from one observer's perspective or (x', t') from another observer's perspective.

This diagram directly shows you the relativity of simultaneity. Events that lie on a single line parallel to x -axis are simultaneous from the frame of reference of the stationary observer and those which lie on any one line parallel to the x' -axis are simultaneous to the moving observer. Clearly you see how the list is not the same for the two observers.

Another interesting thing you see is the trajectory for a beam of light. The beam of light is equally inclined to both pairs of axes – (x, t) and (x', t') . This would imply that the ratio $\frac{\Delta x_{\text{light}}}{\Delta t} = \frac{\Delta x'_{\text{light}}}{\Delta t'} = 1$. Hence both frames would measure the same speed for light.

As the velocity of the moving observer gets closer and closer to the speed of light, the two axes tilt more and more towards the $t = x$ line inclined at 45° to both axes. Finally at the speed of light, both x' -axis and t' -axis superimpose and we get the trajectory of light itself.

Here, we have the primed frame plotted in grid of the unprimed frame. If we were to do the opposite and plot the unprimed frame in the coordinate grids of the primed frame, we would still get a similar diagram with the axes interchanged and flipped along

the vertical t' -axis. Angle between x -axis and t -axis would be acute (not obtuse) from the frame of reference of the moving frame. This confirms once and for all that each observer can rightfully claim to be at rest from their frame of reference.

9.2 Lorentz Transformation as an exotic rotation of space-time diagrams

The question we are trying to answer in this section is whether we can describe mathematically the rotation of space-time diagram that takes us from one set of observers to another.

I first define a variable φ called rapidity, such that

$$e^\varphi = \sqrt{\frac{1+v/c}{1-v/c}} \quad \text{and} \quad e^{-\varphi} = \sqrt{\frac{1-v/c}{1+v/c}}$$

From this, we observe that the Lorentz factor can be written as

$$\gamma = \frac{e^\varphi + e^{-\varphi}}{2} = \cosh(\varphi)$$

The corresponding hyperbolic sine will give us

$$\sinh(\varphi) = \frac{e^\varphi - e^{-\varphi}}{2} = \left(\frac{v}{c}\right) \gamma$$

and the corresponding hyperbolic tangent would be

$$\tanh(\varphi) = \frac{\sinh(\varphi)}{\cosh(\varphi)} = \frac{v}{c}$$

The Lorentz Transformation equations in terms of this sinh and cosh can then be derived as

$$\begin{array}{l|l} \begin{array}{l} x' = \gamma(x - vt) \\ \Rightarrow x' = \gamma x - \gamma \left(\frac{v}{c}\right) ct \\ \Rightarrow x' = x \cosh(\varphi) - ct \sinh(\varphi) \end{array} & \begin{array}{l} t' = \gamma \left(t - \frac{vx}{c^2}\right) \\ \Rightarrow t' = \gamma t - \gamma \left(\frac{v}{c}\right) \frac{x}{c} \\ \Rightarrow t' = t \cosh(\varphi) - \frac{x}{c} \sinh(\varphi) \\ \Rightarrow ct' = ct \cosh(\varphi) - x \sinh(\varphi) \end{array} \end{array}$$

In matrix form, this can be written as

$$\begin{bmatrix} ct' \\ x' \end{bmatrix} = \begin{bmatrix} \cosh(\varphi) & -\sinh(\varphi) \\ -\sinh(\varphi) & \cosh(\varphi) \end{bmatrix} \begin{bmatrix} ct \\ x \end{bmatrix}$$

Note how this has a similar structure to the result we got under transformation for rotated coordinates (9). We can then have a general matrix transformation for 4 dimensional space-time (3 spatial + 1 time) as follows.

$$\begin{bmatrix} ct' \\ x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cosh(\varphi) & -\sinh(\varphi) & 0 & 0 \\ -\sinh(\varphi) & \cosh(\varphi) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} ct \\ x \\ y \\ z \end{bmatrix}$$

So given a velocity v , we can get compute $\tanh(\varphi)$, from that obtain φ , plug that into $\cosh(\varphi)$ and $\sinh(\varphi)$ and do the transformation without ever needing to compute Lorentz factor. Tedious? yes! But that's not the point. The point is that we can geometrically do the transformation without going the algebraic path.

But if you do insist on finding it the Lorentz transformation way but using a 4-vector structure like this, we still do have have

$$\begin{bmatrix} ct' \\ x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma v/c & 0 & 0 \\ -\gamma v/c & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} ct \\ x \\ y \\ z \end{bmatrix}$$

Lorentz Transformation Matrix Example - Rocket's Rapidity

A rocket is moving northward with rapidity φ_v and fires a projectile forward with rapidity φ_w relative to the rocket. What is the rapidity of the projectile from our frame?

Answer: Let the velocity (in units where $c = 1$) of the rocket be v and that of the projectile with respect to the rocket be w . If w' be the velocity of the projectile from our frame. Then, by law of addition of velocities,

$$w' = \frac{v + w}{1 + vw}$$

Since $\tanh(\varphi_v) = v/c$, v can be written as $\tanh(\varphi_v)$ and w can be written as $\tanh(\varphi_w)$ in units where $c = 1$.

$$\begin{aligned} \Rightarrow \tanh(\varphi_{w'}) &= \frac{\tanh(\varphi_v) + \tanh(\varphi_w)}{1 + \tanh(\varphi_v) \tanh(\varphi_w)} \\ \Rightarrow \tanh(\varphi_{w'}) &= \tanh(\varphi_v + \varphi_w) \\ \Rightarrow \varphi_{w'} &= \varphi_v + \varphi_w \end{aligned}$$

[ps: Finally a quantity that adds up algebraically. Newton would be proud.]

Sometimes while drawing space-time diagrams, we use ct instead of t for the time-axis. This is to make sure that the light-cone is still a 45° inclined cone even if we use SI units or any other units where $c \neq 1$.

10 Invariants

So far in Special Relativity, we have come across the quantities that two observers who are in relative motion do not agree upon. They don't agree on when and where an event happens, synchronized clocks and relative velocity of a third observer. It is equally interesting and important to learn about the things that all observers do agree upon. These quantities are called 'invariants'

For example, different cartesian spatial coordinate systems will agree upon the distance between two points because distance is a spatial invariant. In the case of rotation of coordinate grids, origin can act as a fixed point. Then the distance from the origin to any point in space is the same for all rotated frames. Mathematically, $\sqrt{x^2 + y^2} = \sqrt{x'^2 + y'^2}$.

Firstly, all observers must agree that the speed of light in vacuum is $c = 299,792,458\text{m/s}$. This is where we began our discussion of Special Relativity. We shall see two more of such invariants below and one towards the end of this book.

10.1 Space-time interval between events - A space-time invariant

Now we claim that there exist a *space-time invariant* such that different space-time coordinate systems will agree on 'space-time distance' or 'interval'. Mathematically, this is as follows

$$\boxed{-(ct)^2 + (x)^2 = -(ct')^2 + (x')^2}$$

proof: The proof the above result can be obtained from Lorentz transformations.

$$\begin{aligned} (x')^2 - (ct')^2 &= \gamma^2(x - vt)^2 - c^2\gamma^2\left(t - \frac{vx}{c^2}\right)^2 \\ &= \gamma^2\left((x^2 + v^2t^2 - 2xvt) - c^2\left(t^2 + \frac{v^2x^2}{c^4} - \frac{2vxt}{c^2}\right)\right) \\ &= \gamma^2\left(x^2 - c^2t^2 + v^2t^2 - \frac{v^2x^2}{c^2}\right) \\ &= \gamma^2\left(x^2 - c^2t^2 - \frac{v^2}{c^2}(x^2 - c^2t^2)\right) \\ &= \frac{1}{1 - \frac{v^2}{c^2}}\left((x^2 - c^2t^2)\left(1 - \frac{v^2}{c^2}\right)\right) \\ \implies (x')^2 - (ct')^2 &= (x)^2 - (ct)^2 \end{aligned}$$

More generally, if I have two events (x_1, t_1) and (x_2, t_2) in one frame and (x'_1, t'_1) and (x'_2, t'_2) in another frame, then their space-time interval $(\Delta x, \Delta t)$ and $(\Delta x', \Delta t')$ follow

$$\boxed{-(c\Delta t)^2 + (\Delta x)^2 = -(c\Delta t')^2 + (\Delta x')^2}$$

Invariants Example - Space travel As you travel at constant velocity away from Earth, your ship's clock shows that 24 years have elapsed. According to a different set of space-time coordinates, 26 years have elapsed since you departed Earth. According to the latter coordinates, how many light years must you have traveled?

Answer: From my frame, my position is still at my origin. Therefore, $\Delta x' = 0$.

$$\begin{aligned}
 -(c\Delta t')^2 &= -(c\Delta t)^2 + (\Delta x)^2 \\
 \implies (\Delta x)^2 &= c^2(\Delta t^2 - \Delta t'^2) \\
 \implies (\Delta x)^2 &= 1 \frac{[\text{light-year}]^2}{[\text{year}]^2} ((26[\text{year}])^2 - (24[\text{year}])^2) \\
 \implies (\Delta x)^2 &= 100[\text{light-years}]^2 \\
 \implies \Delta x &= 10 \text{ light-years}
 \end{aligned}$$

10.2 Cause & Effect - A space-time invariant

In this subsection, we will examine whether all observers would agree on causality - can one event-A be the cause for an event-B?.

setup : Lets say Alice and Bob are standing on the ground. Alice throws a Ball to Bob and Bob then catches it. From the ground frame, both these events happen 10 seconds apart. For a moving frame however, the interval between these two events need not be 10 seconds and asynchronous nature of clocks also come into play. But is it necessary that in that frame, the event of Bob catching the ball comes after the event of Alice throwing it? In other words, is there a frame where the order of events seem to change - Bob catches the ball before Alice throws it?

Let t_{Ball} be the time at which Alice throws the ball from the ground perspective and t_{Hand} be the time at which Bob catches the ball, again from the ground frame. Similarly t'_{Ball} and t'_{Hand} are also defined, but for the moving frame. From the ground perspective, the ball reaches Bob's hand after it was thrown by Alice. Therefore, $\Delta t = t_{\text{Hand}} - t_{\text{Ball}} > 0$. I also define $\Delta t' = t'_{\text{Hand}} - t'_{\text{Ball}}$ which I don't yet know whether is positive or negative. The temporal order of events from the moving frame can be reversed by showing that $\Delta t'$ can be negative (that is, opposite to that of Δt). By Lorentz transformation,

$$\Delta t' = \gamma \left(\Delta t - \frac{v_{\text{frame}} \Delta x}{c^2} \right)$$

We want to see if we can force $\Delta t < (v_{\text{frame}} \Delta x)/c^2$ so that $\Delta t' < 0$. Rearranging, we get

$$\begin{aligned}
 \frac{\Delta x}{\Delta t} \cdot v_{\text{frame}} &> c^2 \\
 \implies v_{\text{ball}} \cdot v_{\text{frame}} &> c^2
 \end{aligned}$$

For this to be true, atleast one of them should be more than c . Hence, the fact that temporal order cannot be reversed is proved by contradiction.

An easier way to visualize this problem is using space-time diagrams. Alice of course throws the ball at a speed lower than the speed of light. Hence the trajectory for the path of ball through space-time must be more inclined to t -axis than light.

From the ground frame, we see that $t_{\text{Hand}} > t_{\text{Ball}}$. This is why the ball appears to be caught after it was thrown (from the ground frame). The same idea applies to the moving frame as well. If t'_{Ball} and t'_{Hand} lie on different time lines according to the moving observer, these events would not occur simultaneously and also if $t'_{\text{Hand}} > t'_{\text{Ball}}$, the event of catching the ball happens after the event of throwing the ball.

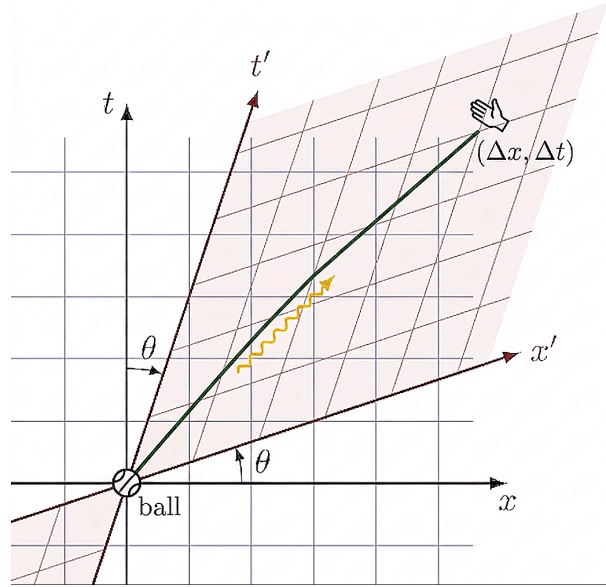


Figure 19: The ball logo is the event of throwing the ball and the hand logo is the event of catching the ball.

To find a contradiction to this temporal order of events, we must draw a space-time diagram where $t'_{\text{Hand}} < t'_{\text{Ball}}$. For that to exist, there must be a space-time diagram where the ball's trajectory goes back in time on the t' axis, and for this, it should have a slope less than the slope of the lines where t' is constant (ie. same as the slope of x').

Slope of ball's trajectory (from ground perspective) = $1/v_{\text{ball}}$. Meanwhile the slope of the x' -axis (in units where $c = 1$) is v_{frame} . For slope of ball to be less than that of x' -axis,

$$\frac{1}{v_{\text{ball}}} < v_{\text{frame}}$$

$$v_{\text{Ball}} \cdot v_{\text{frame}} > 1$$

Again, for this to be true, atleast one of them should be more than $c = 1$. Hence there doesn't exist a valid space-time diagram where the trajectory of the ball is in the $-t'$ direction and subsequently, there doesn't exist a frame where $t'_{\text{Hand}} < t'_{\text{Ball}}$. Temporal Order is thus preserved.

If Δx is the distance traveled by the ball, Δt be the time for which the ball travels (both

from the ground frame),

$$v_{\text{ball}} = \frac{\Delta x}{\Delta t} < c$$

$$\implies (\Delta x)^2 < (c\Delta t)^2$$

From this the ground frame observers would compute the invariant space-time interval as

$$\boxed{-(c\Delta t)^2 + (\Delta x)^2 < 0}$$

Since space-time interval must be the same for all, if one observer says that this quantity is negative, all observers will agree that this quantity is negative.

Hence, the sign of the invariant interval can tell us more about the nature of two events $-(x_1, t_1)$ and (x_2, t_2) . Defining $\Delta x = x_2 - x_1$ and $\Delta t = t_2 - t_1$, we observe 3 cases :-

$$1. -(c\Delta t)^2 + (\Delta x)^2 < 0$$

If this invariant interval is negative, we call them ‘timelike separated’ events (because there exist a frame such that the events will take place at the same place but at different times). Events that are timelike separated can be *causally connected*. Signals going from (x_1, t_1) to (x_2, t_2) travel at a speed less than the speed of light. Pictorially, this refers to when (x_2, t_2) lies inside the light-cone having (x_1, t_1) as origin.

From the inertial frame where the the events appear timelike separated, the invariant interval is simply $-c^2\Delta t'^2$

$$2. -(c\Delta t)^2 + (\Delta x)^2 = 0$$

If this invariant interval is zero, we call them ‘lightlike separated’ events. Events that are lightlike separated can be *causally connected* if the signals going from (x_1, t_1) to (x_2, t_2) travel at the the speed of light. Pictorially, this refers to when (x_2, t_2) lies on the curved surface of the light-cone with (x_1, t_1) as origin.

$$3. -(c\Delta t)^2 + (\Delta x)^2 > 0$$

If the invariant interval is positive, we call them ‘spacelike separated’ (because there exist a frame such that the events will take place at the same time but at different locations). Events that are spacelike separated are *causally disconnected*. Signals going from (x_1, t_1) to (x_2, t_2) has to travel at a speed more than the speed of light. Pictorially, this refers to when (x_2, t_2) lies outside the light-cone having (x_1, t_1) as origin.

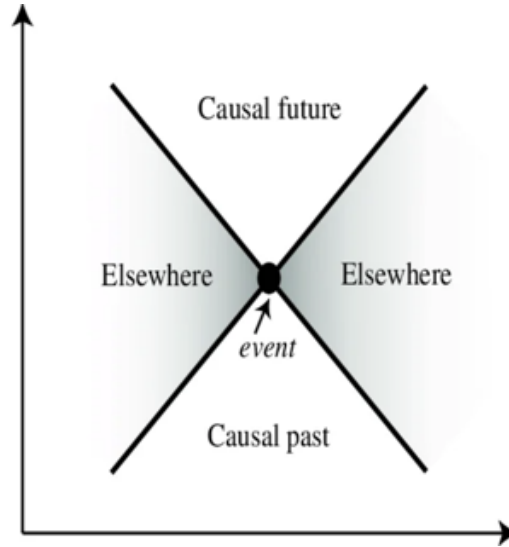
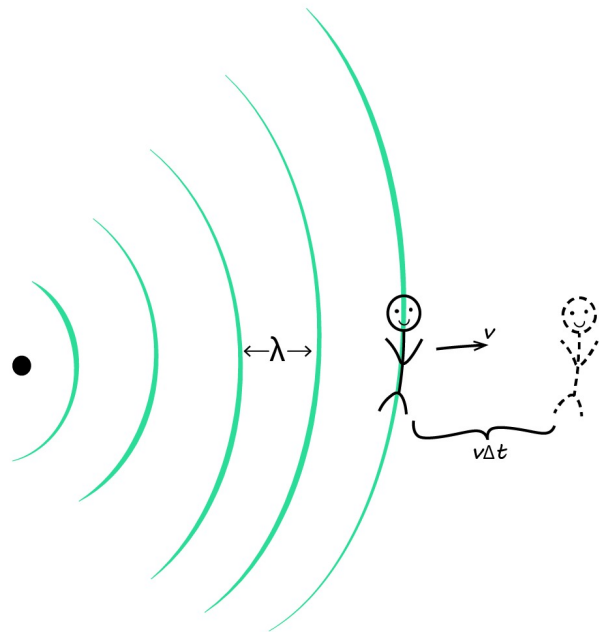


Figure 20: Causal past refers to all those events which can be the cause of “the event” and causal future refers to all those events which can be caused due to “the event”

11 Relativistic Doppler Effect

The relativistic Doppler effect is similar to the every-day Doppler effect that we experience with sound waves. When the source and receiver of a sound wave has relative motion between them, the receiver receives a different frequency of the signal that what was emitted. If the source and receiver are moving away from each other, the frequency of light received is lower and we say that the light is *red-shifted*. Meanwhile if the source and receiver is moving towards one another, the frequency of received light increases and we say that the light gets *blue-shifted*.

setup: We have Alex and Bob on the ground. Alex has a light source that emits light of a particular frequency - ν_0 . Bob is free to either run away from Alex or run towards him. The frequency of the light that Bob receives is ν' . For our first case, let's assume that Bob is running away from Alex. We also have to remember that the speed of light can be written as $c = \nu \cdot \lambda$. Although Alice and Bob may disagree on ν and λ , they will agree that the product will equal the same c .



Since Bob is running away from the light source, the light has to travel an additional $v\Delta t$

to reach him. To find the time difference between subsequent crests – Δt , we write

$$\begin{aligned} c\Delta t &= \lambda + v\Delta t \\ \implies \Delta t &= \frac{\lambda}{c - v} \end{aligned}$$

Notice that Δt is the time interval that we say Bob experiences between each crest. However, remember that Bob is in motion and hence his clocks tick off time more slowly. Thus, the time interval that Bob would measure is actually $\Delta t' = \Delta t/\gamma$

$$\begin{aligned} \Delta t' &= \frac{\lambda}{c - v} \sqrt{1 - v^2/c^2} \\ \implies \Delta t' &= \frac{\lambda}{c(1 - \frac{v}{c})} \sqrt{\left(1 - \frac{v}{c}\right) \left(1 + \frac{v}{c}\right)} \\ \implies \Delta t' &= \frac{\lambda}{c} \sqrt{\frac{1 + v/c}{1 - v/c}} = \frac{1}{\nu_0} \sqrt{\frac{1 + v/c}{1 - v/c}} \end{aligned}$$

Remember that $\Delta t'$ is the time interval between two successive crests from Bob's perspective which is nothing but the observed Time Period. The reciprocal of that is the observed frequency ν'

$$\boxed{\nu' = \nu_0 \sqrt{\frac{1 - v/c}{1 + v/c}}}$$

The case where the observer is running towards the source can also be derived in a similar fashion and one will get a result where the sign of v is replaced in the formula.

One might also realize that unlike sound waves where both v_{observer} and v_{source} plays a role in calculating the frequency change due to Doppler effect, for light we only care about the relative velocity v between the source and observer. This is because sound travels through a medium (air) and therefore identifying the stationary frame is important meanwhile light doesn't move through any such medium (like the *luminiferous ether*) and hence it doesn't matter what the stationary frame is.

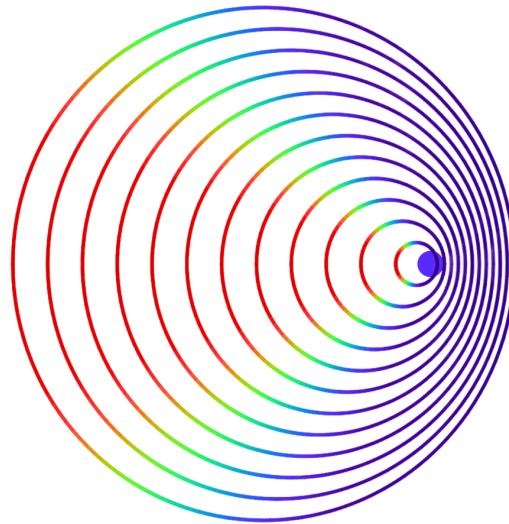


Figure 21: Relativistic Doppler Red-shift and Blue-shift in action

12 Paradoxes in Special Relativity

There are actually no paradoxes in special relativity. If there were, it would imply that the theory is inconsistent. The paradoxes that follow are nothing but seeming paradoxes. It usually happens when we think there is no way to reconcile the different views of different observers. However careful examination of the situation always prove to be otherwise and we always find a logical and coherent explanation.

12.1 Pole in the Barn Paradox

setup: For the pole in the barn paradox we have a 15ft pole, a 10ft barn (both measured at rest) and our earlier two characters – Alice and Bob. When at rest, the 15ft pole will not fit inside a 10ft barn. But lets say Bob is running with the pole towards the barn with $v_{\text{pole}} = 12/13 c$. From the perspective of Alex on the ground, the Lorentz contracted length of the pole would be $15/\gamma = 5.77\text{ft}$ and it could fit inside the still 10ft long barn.

However, from the perspective of Bob running with the pole, the barn's length would have Lorentz contracted to $10/\gamma = 3.8\text{ft}$. Hence, from his perspective, the pole which is 15ft long will not fit inside a 3.8ft barn.

Now we want to know – Who is right?

As we guessed, both are right. To say something is right, is to say one observer can make total sense of how the other observer came to his/her seemingly bizarre conclusions without it contradicting their own conclusions. How does the Alex on the ground conclude that the Bob's conclusions are valid and vice versa?

From Alex's perspective clock on the back of the pole is ahead in time compared to the clock at the front. So Alex would say that when Bob measured the positions of the pole, it was not simultaneous. He would have measured the position of the back of the pole first, let the pole move out of the barn then measure the position of the front and thereby conclude that pole doesn't fit inside the barn.

From Bob's perspective, the reverse would have happened. Bob would say that Alex first measured the position of the front of the pole while it was inside the barn, let the pole move a little and then measure the position of the rear of the pole and hence come to the conclusion that both the front and back of pole simultaneously fit inside the barn at a given instant.

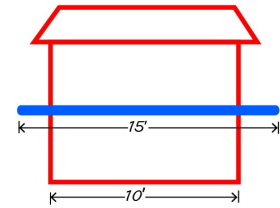


Figure 22: The pole and barn at rest

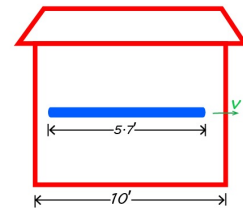


Figure 23: Observed by Alex from the ground

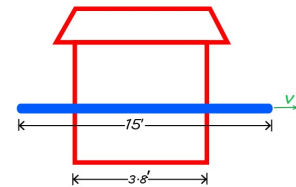


Figure 24: Observed by Bob from the moving pole's perspective

12.1.1 What if we lock the doors?

The thought process for this argument goes like this. What if when the pole fits inside the barn from Alex's perspective on the ground, we close the barn doors? If Alex's perspective of 'the rod fits inside the barn' is valid, then the pole should fit inside the barn's closed doors atleast for sometime. If that's the case, after the rod hits the unmovable doors and stops, it should entirely be inside the barn.

The problems is when Bob who is running with the pole sees what is happening. From his perspective, the pole never fitted inside the barn in the first place. So there cannot be an instant where the barn doors are closed and the entire pole is inside the barn.

So who is right?

Based on earlier trends, you might expect me to say "Both are right!" Good guess but better luck next time. The reader must take one thing to heart. *Relativity is weird, yes.. but its not that weird.* Sometimes both are right just cannot work. Because I can attach side consequences to the dilemma to make it more contrasting and at some point the two perspectives cannot be explained coherently by each other and we would need something like a many-worlds interpretation and so on. Let's say that prior to the experiment, a deal was struck: if the pole doesn't fit inside the closed doors (as Bob insists), Alex and Bob will hunt me down and murder me, leaving this document forever unfinished. If the pole does fit (as Alex insists), I get to live. If we come to a conclusion that both perspectives are right, then that would imply that from one person's perspective, I was murdered and from the other person's perspective, I was not. **More importantly**, from Alex's perspective, this document would be completed in a few weeks but from Bob's perspective, the paper got discontinued in blood.

Hence, no.. both Alex and Bob cannot be right at the same time. In fact it is Alex who is right... The rod can indeed be fit inside a barn with closed unmovable door. To explain this from Bob's perspective, we must invoke the fact that the rod compresses upon impact. It is crucial to note that this compression is not the result of the material property but because of the fact that information cannot be sent faster than the speed of light. Let us take the theoretically most rigid rod possible. When the front of the rod hits the unmovable door of the barn, the front comes to rest at that instant meanwhile the rear end of the pole keeps moving forward until the information to stop reaches the rear end.

In our case, this time interval is 15 nanoseconds from Bob's perspective while 5.3 nanoseconds from Alex's perspective. Hence from both people's perspective, the rear end of pole keeps moving forward even after the front has stopped and this leads to compression of a perfectly rigid rod.

This explains why Alex's prediction is right and I get to live to finish this document.

12.2 The Twin Paradox

The twin paradox is the most famous paradoxes in special relativity. In one way, this paradox uses the argument of relativity that 'Each observer can claim that he/she is in rest and it is the rest of the world moving' against special relativity itself.

setup: We have two characters – Alice and Bob happen to be twins. One day Alice gets on a spaceship and flies off at an incredibly high constant velocity to space while Bob stays on earth. From Bob’s perspective, Alice’s clocks are ticking off time very slowly so that when she comes back to earth, he will say that Alice has aged less than he has. The entire situation is reversed from Alice’s perspective. She would say that it was Bob who was moving and hence his clocks ticked off time more slowly and hence she would be older than her twin brother Bob when they meet again.

Someone here is definitely right and someone here is definitely wrong. I’ll first give the answer then explain how I got there. It is Alice who is younger than Bob.

To put the explanation out bluntly, the equivalence principle of relativity can only be applied when both observers are in an inertial frame of reference throughout the journey. But we know from here that this would not be the case. In our thought experiment, Alice goes away into space, turns around and comes back to earth. That turning around and coming back requires acceleration and deceleration and hence she is not in an inertial frame anymore when that happens. If both observers were to maintain their inertial state of motion, then neither of them will ever come in contact again to find out who is older. In our case, Bob however has been a perfect inertial frame throughout the experiment and hence his observation is valid – Alice is indeed younger than Bob.

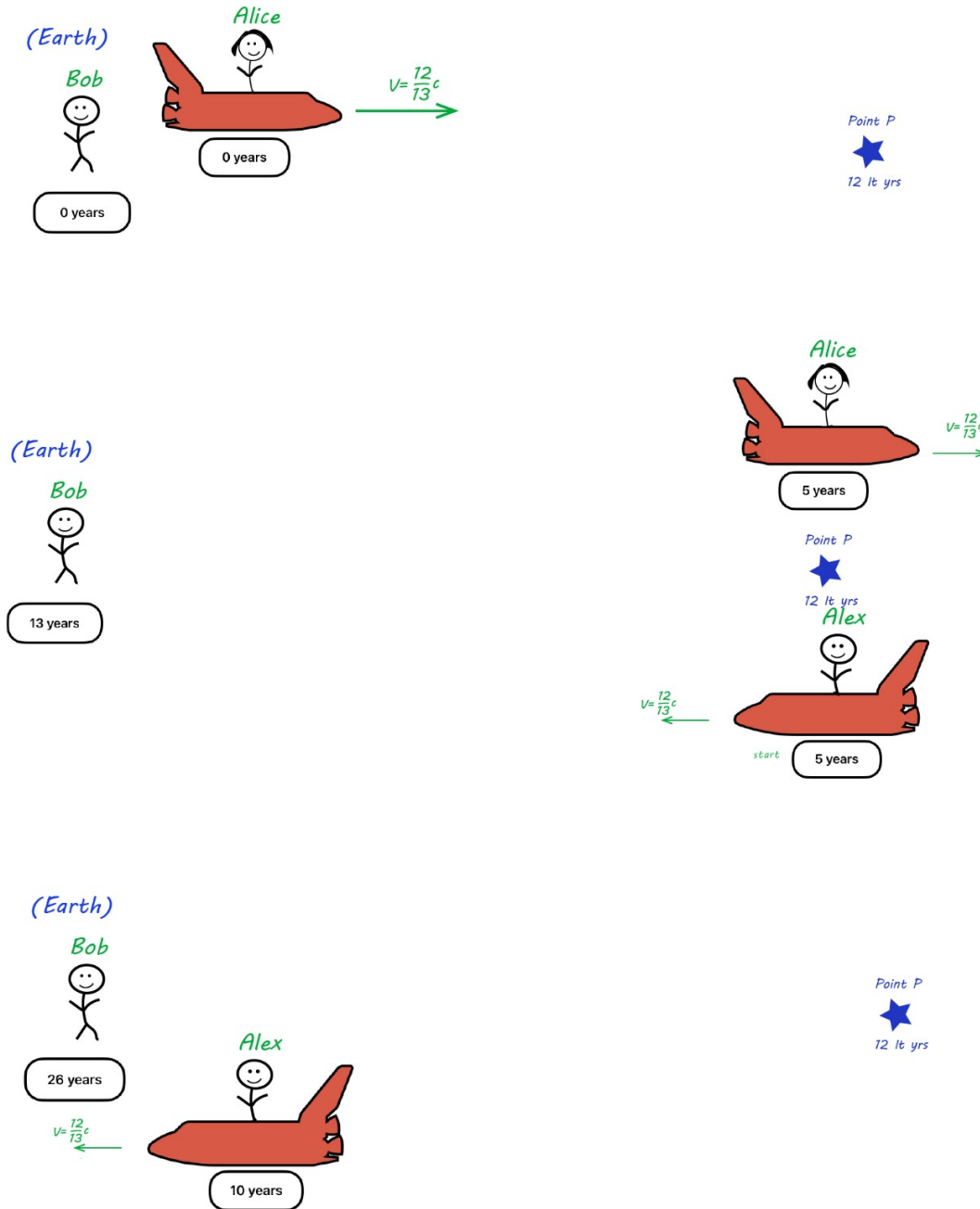
12.2.1 Twin Paradox without Acceleration

The biggest enemy to the paradox was the fact that Alice underwent acceleration and therefore was no longer an inertial where we could safely apply special relativity. What if we could find a work-around to this? That is, have no acceleration in the entire thought experiment. How do we set the thought experiment that way? We invoke a third character – Alex who is moving towards earth as Alice is moving away from it. Here’s how it goes

setup: Alice gets on her spaceship and flies off at a constant velocity of say $12/13 c$ towards right. After a point- P , which is 12 light-years away, Alex who is moving left (towards earth) at the same speed passes Alice. At the instant where Alex passes Alice, he notes down the time reading on her clock and also starts his clock at that instant. Upon reaching earth, he meets up with Bob who takes the reading off Alice’s clock when they passed adds that time to the time that elapsed on Bob’s clock till when he reached earth to get the total time elapsed for a journey out into space and back without the accelerations of turning around and coming back.

From Bob’s perspective, both Alice and Alex take 13 years each to cover 12 light years traveling at $v = \frac{12}{13}c$ and hence he will age $13 + 13 = 26$ years. Bob is aware that both Alice and Alex’s clocks tick off $\gamma = 13/5$ times slower. so each of their clocks should only read 5 years. Bob then comes to the conclusion since Alex here is equivalent to Alice going and coming back without acceleration, that she would only have aged 10 years.

What Bob says would have happened

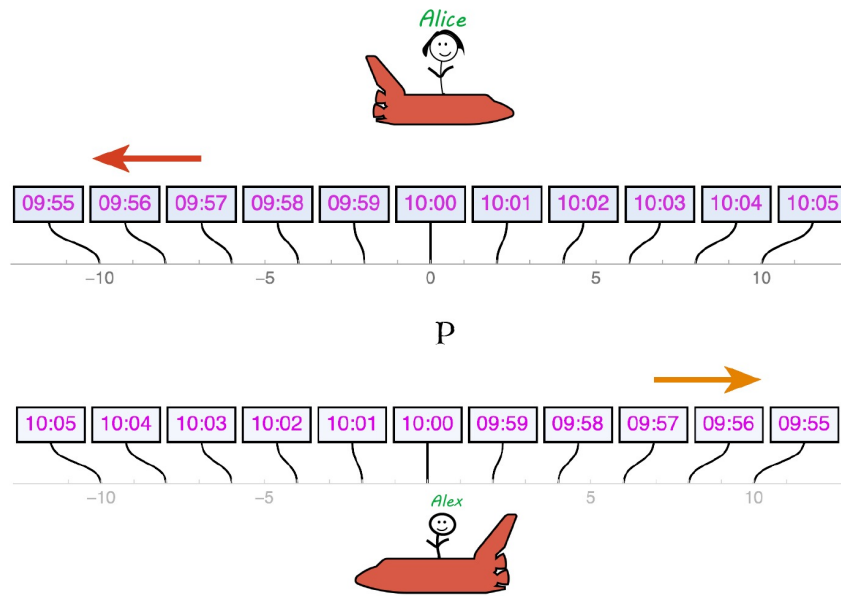


From here on things get interesting. Now let's analyze this situation from Alice and Alex's perspective. Alice after reaching point- P notes that her clock reads 5 years. It makes perfect sense to her because she didn't travel 12 light-years. She traveled a Lorentz contracted $60/13$ light-years. However if she were to comment on Bob's clock now, she would say that Bob was in motion with respect to her and hence his clocks should have ticked off less time than her.

$$\Delta t_{\text{Bob w.r.t Alice}} = \frac{60}{13} \times \frac{1}{\gamma} = \frac{25}{13} \approx 1.923 \text{ years}$$

If that were the case, then by symmetry, Alex would also claim that only 1.923 years would have passed on Bob's clock because he too will claim to be at rest and argue that Bob's clock tick off less than 5 years that passed on Alex's clock. We add these numbers up and see that only $\frac{50}{13} \approx 3.85$ years should pass on Bob's clock from Alice and Alex's perspective for the entire journey.

One might assume that we have reached a clear contradiction. But on close inspection we realize that we have made a mistake. We assumed that the reading on Bob's clock will be the same for both Alice and Alex when they meet at point- P . However, this will not be the case. Why? Because of the asynchronous nature of clocks in motion. Since Alice and Alex are moving in opposite directions, they will not agree that all of Bob's clocks show the same time (even if the rate at which time elapses on Bob's clocks are the same for both Alice and Alex).



As you can see above, the only clock on Bob's frame that they both agree on is the one at point- P as they pass. Thus we should first find the time on Bob's clock at Point- P at that instant.

From Alice's perspective, the time on Bob's clock at point- P is ahead of the time on Bob's clock on earth by a factor of vD/c^2 . That is...

$$\begin{aligned}
 t_{\text{Bob (at } P) \text{ wrt Alice}} &= t_{\text{Bob (on earth) wrt Alice}} + \frac{vD}{c^2} \\
 &= \frac{25}{13} + \frac{\frac{12}{13} \times 12}{1^2} \\
 &= \frac{25}{13} + \frac{144}{13} = 13 \text{ years}
 \end{aligned}$$

Now, as we mentioned before, Alex would agree that the time on Bob's clock at point- P is 13 years. Now however, for him the time that reads on Bob's clock (on earth) is ahead

of this time by a factor of vD/c^2

$$\begin{aligned}
 t_{\text{Bob (on earth) wrt Alex}} &= t_{\text{Bob (at } P \text{) wrt Alex}} + \frac{vD}{c^2} \\
 &= 13 + \frac{\frac{12}{13} \times 12}{1^2} \\
 &= 13 + \frac{144}{13} \\
 &= \frac{313}{13} \approx 24.077 \text{ years}
 \end{aligned}$$

As calculated before, on the journey back to earth, Alex would say that only 1.923 years would pass on Bob's clock because his clocks tick off time more slowly. We add this to the 24.077 years we got now to get

$$\frac{313}{13} \text{ years} + \frac{25}{13} \text{ years} = 26 \text{ years}$$

Thus, when Alex compares his age to Bob upon reaching earth, he indeed finds that Bob has aged more than him (and Alice). The paradox is thus resolved!

12.2.2 Space-time diagram for the twin paradox (without acceleration)

By drawing the space-time diagram our previous explanation becomes clear. Alice (in red) moves outwards into space to point- P where $x = 12$ light-years and Alex (in green) comes back to $x = 0$. At point- P , we draw the now slices for Alice and Alex and observe that they both do not agree on what the time shown on Bob's clock on earth is at that instant. But they do agree on the time at point- P . Extending the now slice back, we see each of their argument that only 25/13 years have passed for Bob on earth during each of their motions to and from point- P . A large section of earth's time gets *skipped* when we go from Alice's perspective to Alex's perspective. When we add all these up, we do see that it adds up to 26 years as expected.

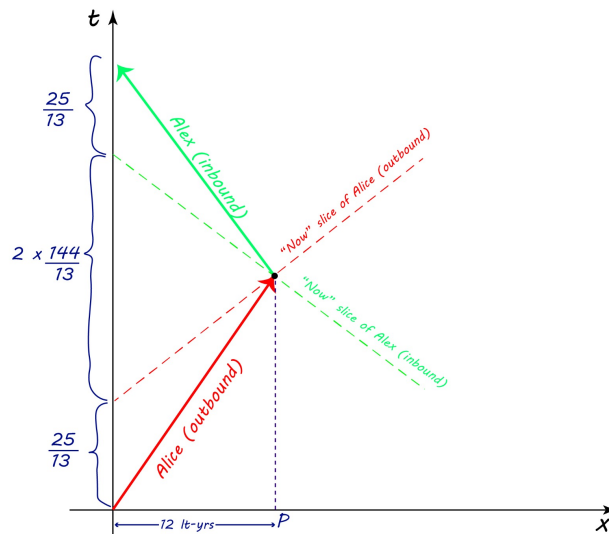


Figure 25: Space-time diagram for the twin paradox (without acceleration).

12.2.3 What if the twins communicate?

Remember earlier that we introduced coordinates for space and time to disregard the light travel time, the post processing required and all? Here we're going to bring that back. We'll try to see what each of the twins would see if they tried to communicate with one another during the journey. We know the answer is going to be that Bob will be older but where will the asymmetry arise?

setup: Just as before, Alice gets on a rocket and flies off into space. Only this time, Bob has a laser pointer with him that he uses to send Alice a signal once every year. From Alice's perspective, receiving a signal pulse of this sort means Bob has aged one year back on earth. By counting the number of pulses she gets, she will figure out how many years Bob would have aged by the time she returns back. Similarly, Alice sends out a laser signal once every year so that Bob can find out how many years has passed on her ship. Now all we need to do is count the number of light pulses each say they receive so that we can find out how old the other person is. Or in other words rate at which an observer receives flashes from another observer is the rate at which he would say that the other observer is aging.

Now, let us first analyze what Bob say will happen. Firstly from his perspective, on the outbound journey of Alice, time runs slower on Alice's clock and therefore she sends out flashes less often than if she were staying still. Also, since she is moving away from Bob the flashes that she sends out take more and more time to reach her and similarly the flashes that Bob sends out to Alice has to travel more distance each subsequent year and hence she receive them less often. On the inbound journey however Alice will be moving towards Bob and hence each light pulse travels less distance than the previous and both Bob and Alice will receive pulses more often. Alice from her perspective will also say the same for how often Bob will send out and receive flashes.

An important thing to notice here is that since it was Alice's act of turning around that completes the experiment, her actions affect her immediately, she starts receiving Bob's light pulses immediately as she turns meanwhile the pulses she send out is only received by Bob later. This would imply that Alice receives more flashes from Bob than Bob receives from Alice making Bob age more than Alice.

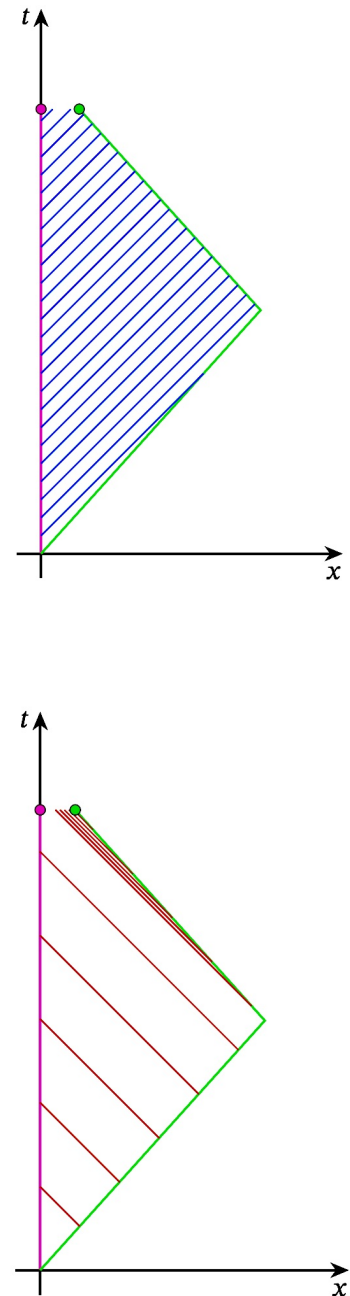


Figure 26: Blue is the flashes sent out by Bob and red is the flashes sent out by Alice.

Remember how we earlier dealt with the relativistic Doppler effect. There we talked about the frequency(color) of light. It is just as relevant here because each signal can be considered as a wave crest. Then calculating the observed rate of the flashes is analogous to the frequency of light observed. This lets us see quantitatively how the communication with laser works out and where the asymmetry arises from. First we need the number of flashes observed by each observer during the outbound and inbound journey.

$$\begin{aligned}
 \nu_{\text{outbound}} &= \nu_0 \sqrt{\frac{1 - v/c}{1 + v/c}} \\
 &= 1 \sqrt{\frac{1 - 12/13}{1 + 12/13}} \\
 &= \frac{1}{5} \text{ flashes per year}
 \end{aligned}$$

$$\begin{aligned}
 \nu_{\text{inbound}} &= \nu_0 \sqrt{\frac{1 + v/c}{1 - v/c}} \\
 &= 1 \sqrt{\frac{1 + 12/13}{1 - 12/13}} \\
 &= 5 \text{ flashes per year}
 \end{aligned}$$

Now let's work out what this means for Alice and Bob.

Alice's perspective

From her perspective, 5 years passed for her from the moment she leaved earth till she reached the turn-around point- P . During this 5 years, she receives

$$5 \text{ years} \times \frac{1}{5} \text{ flashes per year} = 1 \text{ flash}$$

During the Inbound Journey, 5 more years elapses for her and she gets

$$5 \text{ years} \times 5 \text{ flashes per year} = 25 \text{ flashes}$$

Hence, during the entire journey, she gets $1 + 25 = 26$ flashes in total. For Alice, Bob sends out a flash every time he ages by 1 year. Thus, from Alice's perspective, Bob ages by 26 years during the entire journey.

This once again confirms that the **asymmetry kicks in because it is Alice's act of turning around** that dictates whether the journey is outbound or inbound. For her, the moment she turns around, the journey becomes inbound. But for Bob, it takes an additional 12 more years before he even sees that Alice has turned around.

Bob's perspective

From Bob's perspective, 13 years pass for him from the moment Alice leaves earth till she reaches the turn-around Point- P . But although she turns around at Point- P , that information about her turning doesn't reach Bob until 12 years (because she is 12 light-years away). Therefore, for the first $13 + 12 = 25$ years, Bob's reality dictates that Alice is outbound and hence, during this time, he receives flashes according to the outbound formula

$$25 \text{ years} \times \frac{1}{5} \text{ flashes per year} = 5 \text{ flash}$$

After the signal of her turning around does finally reach him, he gets the flashes according to the inbound formula for only the remaining 1 year.

$$1 \text{ years} \times 5 \text{ flashes per year} = 5 \text{ flashes}$$

Hence, during the entire journey, she gets $5 + 5 = 10$ flashes in total. For Bob, Alice sent out a flash every time she aged by 1 year. Thus, from his perspective, she aged only 10 years during the entire journey.

Relativistic Doppler Effect Example - The Twins Communicate

Imagine that Bob is firing photons toward Alice at a rate of N photons per second. Imagine that Alice is running toward Bob at a speed v .

(i) From Bob's perspective, what is the distance between each of the photons that are shooting toward Alice?

Answer: Bob sends out photons at Alice at the rate of N photons per second. This would mean that the time interval between successive photons is $1/N$. During this time interval, a photon sent out travels a distance of c/N before the next photon is released. Hence distance between successive photons is c/N .

(ii) From Bob's perspective, what is the time interval between Alice receiving one photon and the next?

Answer: Let Δt be the time interval between Alice receiving photons from Bob's perspective. During this Δt , Bob sees Alice move closer to him by an amount $v\Delta t$. Therefore, the distance the photon has to travel – $c\Delta t$ to reach Alice is less than c/N .

$$\begin{aligned} c\Delta t &= \frac{c}{N} - v\Delta t \\ \implies \Delta t &= \frac{c}{N(c+v)} \end{aligned}$$

(iii) From Bob's perspective, how many photons does Alice receive per second?

Answer: Reciprocal of the time interval between receiving photons gives rate.

$$\text{rate} = N \left(1 + \frac{v}{c}\right)$$

(iv) From Alice's perspective, how many photons does Bob send out per second?

Answer: From Alice's perspective, Bob's clock runs slow by a factor of γ . Hence number of photons per second is N/γ

(v) From Alice's perspective, what is the distance between each successive photon that Bob fires?

Answer: From Alice's perspective, photons are sent out at the rate of N/γ photons per second but Bob is moving towards her at a rate of v . Hence each subsequent photon travels $vT' = \gamma v/N$ less. Thus distance traveled by the photon is

$$\begin{aligned} D' &= \frac{\gamma c}{N} - \frac{\gamma v}{N} \\ \implies D' &= \frac{\gamma(c-v)}{N} \end{aligned}$$

(vi) According to Alice what is the time interval between the arrival of successive photons?

Answer: According to Alice, the time interval between arrival of photons is D'/c . That is

$$T' = \frac{\gamma(1 - v/c)}{N}$$

(vii) According to Alice, then, how many photons does she receive per second?

Answer: **Answer:** If Bob says Alice gets photons with a time interval of Δt , Alice would say she gets them at a rate of $\Delta t' = \Delta t/\gamma$

$$\begin{aligned}\Delta t' &= \frac{c}{N(c+v)} \sqrt{1 - v^2/c^2} \\ \Rightarrow \Delta t' &= \frac{1}{N} \sqrt{\frac{1 - v/c}{1 + v/c}} \\ \Rightarrow \text{rate} &= N \sqrt{\frac{1 + v/c}{1 - v/c}}\end{aligned}$$

Twin Paradox Example - The Twins Communicate

For the twin paradox, take Alice's speed to be $v = 12/13 c$ and the distance according to Bob to the turnaround point to be 12 light-years. Assume both Bob and Alice emit a flash each time their own clock shows that a year has elapsed. Based on Alice's observations, what will she conclude about the number of flashes Bob emits during the moment she turned around to head back to Earth?

Answer: From Alice's perspective, it takes 5 years for her to reach the turnaround point. But Bob's clocks tick off slower than her and only $5/\gamma = 1.923$ years would pass for Bob from her perspective. Since Bob sends out only 1 flash per year, only one of Bob's outbound flashes reach her before she turns around. Now after she turns around, her idea of "the present" on Bob's frame changes. By time symmetry (since inbound journey is just outbound journey played in reverse), only 1.923 years passes on Bob's clock according to Alice after she turns around. Thus again only 1 flash is sent out by Bob (from her perspective) after she turns around.

Since Alice receives a total of 26 flashes from Bob during the journey, she would say that the remaining 24 flashes occurred during her turn-around.

13 The Mass-Energy Equivalence

13.1 Implications for Mass

This section deals with the implications that special relativity has on the mass of a body. Specifically we'll see how mass of a body is related to its energy and hence derive Einstein's famous equation $E = mc^2$.

setup: We have two frames – an unprimed frame (S) which is stationary and a primed frame (S') which is moving at a speed v with respect to the unprimed frame. Let us also assume that there is a completely elastic collision between two identical balls taking place from the unprimed frame. From S -frame, the two identical balls are moving towards each other at the same speed u , collides at their common center of mass and rebounds back at the same speed. From the S -frame, the center of mass remains fixed ie. $V_{\text{center of mass}} = 0$. Therefore, it is safe to say that the velocity of the center of mass (from any frame) is the velocity of the unprimed frame.

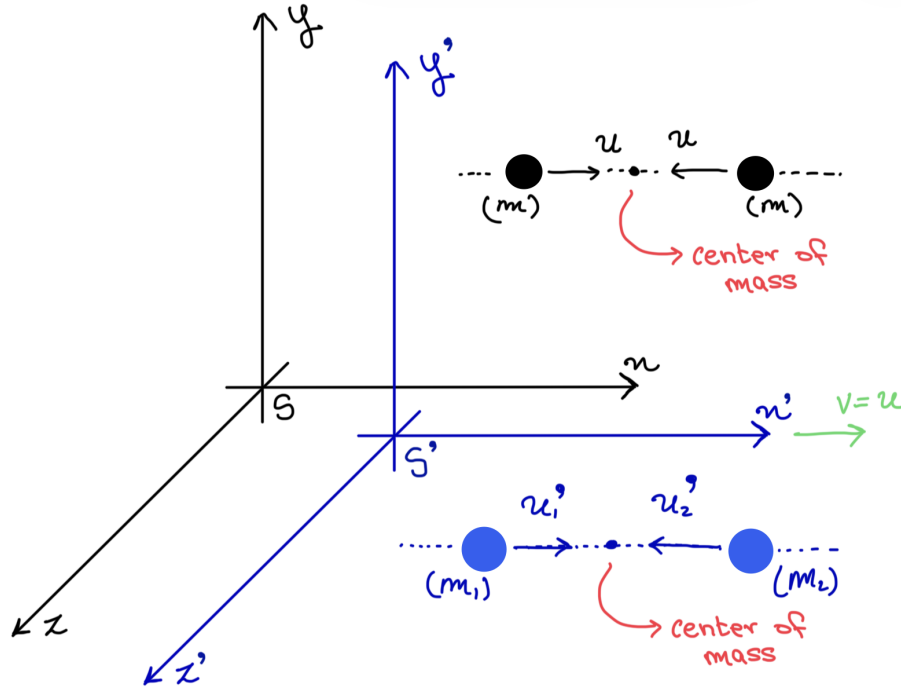


Figure 27: Two frames S and S' moving relative to each other observe the same event of an elastic collision between two balls.

Now we observe the same collision from S' -frame. Technically we can take any arbitrary speed v for the primed frame with respect to the unprimed frame, but for simplicity, if we take $v = u$, it greatly simplifies the calculation since we will then be dealing with two less variables. From the S' -frame, this is what we observe :

Ball-1 (of mass say m_1) is stationary from the frame of S' , meanwhile Ball-2 is moving leftwards at a higher speed u'_2 . We can find u'_2 by applying the relativistic addition of velocities

$$u'_2 = \frac{-u + -v}{1 + \frac{uv}{c^2}} = \frac{-(u + u)}{1 + \frac{u^2}{c^2}} = \frac{-2u}{1 + \frac{u^2}{c^2}}$$

Instead of writing u'_2 in terms of u everywhere in the following steps, it is more relevant to write u in terms of u'_2 . This is because, we as members of frame- S' are more interested in the speed of ball-2 with respect to us than what the actual speeds of the balls or the speed of frame- S is.

$$\begin{aligned}
u'_2 \left(1 + \frac{u^2}{c^2} \right) + 2u &= 0 \\
\implies u'_2(c^2 + u^2) + 2uc^2 &= 0 \\
\implies u'_2u^2 + 2uc^2 + c^2u'_2 &= 0
\end{aligned}$$

We arrive at a Quadratic equation in u that upon solving gets us

$$u = -\frac{c^2}{u'_2} \left(1 \pm \sqrt{1 - \frac{u'^2_2}{c^2}} \right)$$

It is clear from this equation that for very small values of u'_2 , we must maintain the term inside the bracket to be less than 1 rather than greater than 1. Hence, we should take the negative root out of the plus-or-minus

$$u = -\frac{c^2}{u'_2} \left(1 - \sqrt{1 - \frac{u'^2_2}{c^2}} \right)$$

Now, the conservation of linear momentum should still hold in this new frame. I can write the sum of individual momenta as equal to the momentum of the center of mass of the two ball system from frame- S' .

$$\begin{aligned}
m_1u'_1 + m_2u'_2 &= (m_1 + m_2)V'_{\text{center of mass}} \\
(m_1 \cdot 0) + (m_2 \cdot u'_2) &= (m_1 + m_2)V'_{\text{center of mass}}
\end{aligned}$$

Since $V'_{\text{center of mass}}$ is same as the velocity of frame- S with respect to frame- S' , I can write

$$m_2u'_2 = (m_1 + m_2)(-u)$$

I replace the $(-u)$ in terms of u'_2 as

$$\begin{aligned}
m_2u'_2 &= (m_1 + m_2) \left(\frac{c^2}{u'_2} \left(1 - \sqrt{1 - \frac{u'^2_2}{c^2}} \right) \right) \\
m_2 \left(\frac{u'^2_2}{c^2} \right) &= (m_1 + m_2) \left(1 - \sqrt{1 - \frac{u'^2_2}{c^2}} \right) \\
m_2 \left(\frac{u'^2_2}{c^2} \right) &= m_1 \left(1 - \sqrt{1 - \frac{u'^2_2}{c^2}} \right) + m_2 \left(1 - \sqrt{1 - \frac{u'^2_2}{c^2}} \right) \\
m_2 \left(\frac{u'^2_2}{c^2} - 1 + \sqrt{1 - \frac{u'^2_2}{c^2}} \right) &= m_1 \left(1 - \sqrt{1 - \frac{u'^2_2}{c^2}} \right) \\
m_2 \left(- \left(\sqrt{1 - \frac{u'^2_2}{c^2}} \right)^2 + \sqrt{1 - \frac{u'^2_2}{c^2}} \right) &= m_1 \left(1 - \sqrt{1 - \frac{u'^2_2}{c^2}} \right)
\end{aligned}$$

I can take one square root term out of the bracket. I then get

$$m_2 \sqrt{1 - \frac{u_2'^2}{c^2}} \left(1 - \sqrt{1 - \frac{u_2'^2}{c^2}} \right) = m_1 \left(1 - \sqrt{1 - \frac{u_2'^2}{c^2}} \right)$$

Notice that the big term on the left and right side are identical and gets canceled off leaving us with

$$\begin{aligned} \implies m_2 \sqrt{1 - \frac{u_2'^2}{c^2}} &= m_1 \\ \implies m_2 &= \frac{m_1}{\sqrt{1 - u_2'^2/c^2}} \end{aligned}$$

Let us recall term-by-term what this means. We were considering the case of collision between **identical balls**. For us (frame- S'), m_1 refers to the mass of ball-1 which from our frame is stationary. Hence, we can call it the *rest-mass* of the ball. m_2 refers to the mass of ball-2 which from our frame is moving with velocity u_2' .

Therefore, when a body of rest mass m_0 is traveling with speed v , its relativistic mass(m) is given by

$$\boxed{m = \frac{m_0}{\sqrt{1 - v^2/c^2}} = m_0 \gamma}$$

So far in our discussion, I've casually stated or indirectly implied that the speed of light c is the absolute speed limit for our universe. But I never explained why that is so. What stops something from being accelerated by a force to a speed more than c ? The answer of course lies in what you see above – Relativistic mass. In order to accelerate a body by some amount, we need to apply a force that is proportional to the mass of the object. when $v \rightarrow c$, the mass of the object shoots up to infinity implying that an infinite force would be required to produce any small increase in speed. At $v = c$ (practically impossible but say), $m = \infty$ and we would need an infinite force to produce any non-zero acceleration at all.

13.2 Momentum in Special Relativity

Newton's second law is traditionally written as

$$F = ma \quad \text{or} \quad F = \frac{dp}{dt}$$

Where momentum p is written as $p = mv$.

But after being exposed to relativistic mass, we must not treat m as a constant but as a function of velocity. Relativistic momentum is then written as

$$\boxed{p = \gamma m_0 v}$$

A subtle yet interesting thing to appreciate is that using this new definition of momentum, we see that the momentum of a particle along an axis perpendicular to the relative motion

of different frames we observe it from doesn't change. I shall explain it with the following example.

setup: I invoke our earlier system of unprimed S -frame and primed S' -frame. But instead of two balls, I have one ball moving in the y -direction according to frame- S with velocity u . Let us calculate what the momentum of this ball would be from frame- S .

$$p_y = \gamma m_0 u = \frac{m_0 u}{\sqrt{1 - u^2/c^2}}$$

Now, unlike last time, we take a more general relative velocity v between frame- S and frame- S' . From frame- S' , the ball takes a diagonal path through space with some x -component as well as some y -component. Since the x -coordinate of the ball stays fixed to a particular x -coordinate of frame- S , the x -component of velocity (u'_x) that frame- S' observers the ball have is $-v$ but to find the y -component of velocity (u'_y), I must use the relativistic addition of velocities in 3-D. That is,

$$u'_y = \frac{u}{\gamma[v](1 - \frac{v \times 0}{c^2})} = \frac{u}{\gamma[v]}$$

When we multiply that with the relativistic mass $m = \gamma[v'_{\text{ball}}] \cdot m_0$ (where this γ is based on the total velocity of the ball from the S' -frame), we get

$$\begin{aligned} p'_y &= m u'_y = \gamma[v'_{\text{ball}}] \cdot m_0 \left(\frac{u}{\gamma[v]} \right) \\ \Rightarrow p'_y &= \frac{m_0 u \sqrt{1 - v^2/c^2}}{\sqrt{1 - \frac{(u^2 + v^2)}{c^2}}} = \frac{m_0 u \sqrt{1 - v^2/c^2}}{\sqrt{1 - \frac{u^2}{\gamma[v]^2 c^2} - \frac{v^2}{c^2}}} \\ \Rightarrow p'_y &= \frac{m_0 u \sqrt{1 - v^2/c^2}}{\sqrt{1 - \frac{u^2(1 - v^2/c^2)}{c^2} - \frac{v^2}{c^2}}} = \frac{m_0 u \sqrt{1 - v^2/c^2}}{\sqrt{(1 - \frac{v^2}{c^2}) - \frac{u^2}{c^2} (1 - \frac{v^2}{c^2})}} \\ \Rightarrow p'_y &= \frac{m_0 u \sqrt{1 - v^2/c^2}}{\sqrt{(1 - \frac{v^2}{c^2}) (1 - \frac{u^2}{c^2})}} = \frac{m_0 u}{\sqrt{1 - u^2/c^2}} \end{aligned}$$

Thus, $p_y = p'_y$ when the relative motion of two frames S and S' is only along the x -direction or z -direction. Now we see that momentum along perpendicular directions is conserved (or rather **same**).

13.3 Relativistic Work and Kinetic Energy

To obtain an expression for Kinetic energy, we shall first recall the work-energy theorem that states that work done on a system produces a change in its Kinetic Energy. This can be further written as

$$d(KE) = \text{Work Done} = F \cdot dx = \frac{dp}{dt} \cdot dx = v dp$$

But remember, I can write momentum as $p = \gamma m_0 v$. Which implies

$$\begin{aligned}
 p &= \frac{m_0 v}{\sqrt{1 - v^2/c^2}} \\
 \Rightarrow \frac{dp}{dv} &= m_0 \frac{d}{dv} \left(\frac{v}{\sqrt{1 - v^2/c^2}} \right) \\
 &= m_0 \left(\frac{\sqrt{1 - v^2/c^2} - \frac{-2v^2/c^2}{2\sqrt{1 - v^2/c^2}}}{1 - v^2/c^2} \right) \\
 &= m_0 \left(\frac{(1 - v^2/c^2) + v^2/c^2}{(1 - v^2/c^2)^{3/2}} \right) \\
 &= \frac{m_0}{(1 - v^2/c^2)^{3/2}} \\
 \Rightarrow dp &= \frac{m_0}{(1 - v^2/c^2)^{3/2}} dv
 \end{aligned}$$

Integrating the work-energy theorem on both sides, I get

$$\begin{aligned}
 \int_0^{KE} d(KE) &= \int_0^p v dp \\
 \Rightarrow KE &= \int_0^v \frac{m_0 v}{(1 - v^2/c^2)^{3/2}} dv
 \end{aligned}$$

I substitute $v^2 = t$ which implies $v dv = \frac{1}{2} dt$.

$$\begin{aligned}
 \Rightarrow KE &= \frac{m_0}{2} \int_0^{v^2} \frac{1}{(1 - t/c^2)^{3/2}} dt \\
 &= \frac{m_0 c^3}{2} \left[\frac{2}{\sqrt{c^2 - t}} \right]_0^{v^2} \\
 &= m_0 c^3 \left(\frac{1}{c\sqrt{1 - v^2/c^2}} - \frac{1}{c} \right) \\
 &= m_0 c^2 \left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right)
 \end{aligned}$$

Earlier we saw the mass of a body soaring to infinity as the speed of the object gets closer to the speed of light. Now we see that the kinetic energy of a body also soars to infinity (as you might have expected) as we take it to the speed of light.

If we split the brackets and write it as

$$KE = \gamma m_0 c^2 - m_0 c^2$$

where γm_0 is the actual mass possessed by the moving body. Then we see that the total Kinetic energy that a body gains when it is moving at a speed v is nothing but

$$\boxed{KE = (\Delta m) c^2}$$

where Δm is the difference between the relativistic mass and rest mass.

The reader must also appreciate how under everyday conditions where $v \ll c$, we can use the Taylor approximation of Lorentz factor to obtain the familiar Newtonian expression for kinetic energy.

$$KE = m_0 c^2 \left(\left(1 + \frac{1}{2} \frac{v^2}{c^2} + \dots \right) - 1 \right) \approx \frac{1}{2} m_0 v^2$$

13.4 Its finally time for $E = mc^2$

We are now ready to take on $E = mc^2$. We already saw a hint of this in the expression for Kinetic energy of a body. Firstly, we recall that the total energy of a body is the sum of its potential and Kinetic energy. That is,

$$E_{\text{total}} = PE + KE$$

We already have an expression for the KE . We just now need an expression for the other energies. For that, we'll do what we did to derive the relativistic momentum equation – Observe a collision from two different frames. Only this time, instead of a perfectly elastic collision, we analyze a perfectly inelastic collision.

setup: Just as before, we have two frames - S and S' . We have two particles of rest mass m_0 each. From frame- S , the two particles are moving towards each other at speed v , slam into one another, combine to form a larger particle of mass M_0 and stop moving entirely. We may expect M_0 to be the sum of the individual rest masses ($m_0 + m_0 = 2m_0$) because nevertheless the atoms that make up the bigger particle at rest are the same atoms that make up the constituent particles at rest (they just got rearranged). But we shall soon see that this is not the case. From frame- S' moving at v , particle-1 is at rest while particle-2 is moving towards particle-1 at v'_2 . They again combine to form a bigger particle but this bigger particle continues to move with a speed v backwards. v'_2 can be calculated using relativistic addition of velocities as

$$v'_2 = \frac{-v + -v}{1 + v^2/c^2} = \frac{-2v}{1 + v^2/c^2}$$

The momentum of particle-2 from frame- S' is given by

$$\begin{aligned} p'_2 &= \frac{m_0 v'_2}{\sqrt{1 - v'^2_2/c^2}} = \frac{m_0 \left(\frac{-2v}{1+v^2/c^2} \right)}{\sqrt{1 - \frac{(-2v)^2}{c^2(1+v^2/c^2)^2}}} \\ &= \frac{-2m_0 v c^2}{\sqrt{(v^2 + c^2)^2 - 4v^2 c^2}} = \frac{-2m_0 v}{1 - v^2/c^2} \end{aligned}$$

Similarly, the momentum of the combined particle from frame- S' is given by

$$p'_{\text{combined}} = \frac{M_0(-v)}{\sqrt{1 - v^2/c^2}}$$

I can now apply conservation of momentum to this inelastic collision from frame- S' . Before collision, the total momentum of the system was simply just the momentum of particle-2 and after collision, the total momentum of the system is the momentum of the combined particle. That is,

$$\begin{aligned} p'_{\text{combined}} &= p'_2 \\ \Rightarrow \frac{M_0 v}{\sqrt{1 - v^2/c^2}} &= \frac{2m_0 v}{1 - v^2/c^2} \\ \Rightarrow M_0 &= \frac{2m_0}{\sqrt{1 - v^2/c^2}} \end{aligned}$$

Now, I want you to look at this formula very closely, although it might seem like one of those relativistic-mass equations, it is not! M_0 is the **rest-mass** of the combined particle and what this is telling us is that *the **rest-mass** of the combined particle is equal to the sum of the **relativistic masses** of its constituent particles just before collision.*

It is telling us that the kinetic energy of particle-1 and particle-2 as they are coming in is transmuted into some additional rest mass when they coalesce to form a bigger particle. Let us calculate how much bigger that rest-mass is than what we anticipated.

$$\Delta M_0 = \frac{2m_0}{\sqrt{1 - v^2/c^2}} - 2m_0 = 2m_0(\gamma - 1)$$

We also need to calculate the kinetic energy of the particles from frame- S before they combined. It is simply

$$KE_{\text{initial}} = 2m_0 c^2 (\gamma - 1)$$

We can think of this as the change in the kinetic energy being somehow equivalent to the change in mass. $\Delta KE \iff \Delta M_0$ and that relation is of course

$$\Delta KE = (\Delta M_0) c^2$$

The form we obtained here is surprisingly identical to the one we got under the derivation of the relativistic kinetic energy of a particle but the meaning they hold has a subtle difference. Over there, it meant that the kinetic energy a particle gains is equal to the relativistic mass it gains over the rest mass and the whole multiplied by c^2 . Here it means that the change in rest mass that accompanies when making a system from constituent particles is directly proportional to the loss in kinetic energy during the formation.

Now, I want you to really focus on that **additional** rest mass. Focus on it enough to forget about the rest mass that was already there. You must **convince yourself** that this additional rest mass is no different from any other mass in this universe. If I were to cut out this additional rest mass and call it m_0 from now on, it is as if what I holding onto is a frozen form of rest energy. More interestingly, a third person would never be able to tell the difference between the additional rest mass (m_0) I am holding onto and the total mass (M_0) it was cut off from and *who knows whether the mass M_0 was such an additional rest mass from any such collisions in the past.* Therefore, any property that this m_0 shows must also be true for M_0 and all other masses in this universe. Thus all masses at rest possess some rest energy.

If the mass I am holding onto - m_0 was instead moving, it would have an additional kinetic energy of $m_0c^2(\gamma - 1)$. Thus the total energy would become

$$E_{\text{total}} = m_0c^2(\gamma - 1) + m_0c^2 = \gamma m_0c^2$$

Since $m_0\gamma$ is simply the actual mass (m) of the body irrespective of whether it is moving or not. The above expression simplifies to

$$E_{\text{total}} = mc^2$$

beautiful...

14 Transformations for Energy and Momentum

Earlier we dealt with transformation laws for spatial and time coordinates of an event when going from one frame to another. Now we shall deal with the transformation of energy and momentum when changing frames and also arrive at yet another invariant which is in terms of Energy and momentum. For that, first we see how the energy transforms from frame to another

setup: Just as before, we have our frame- S and frame- S' moving with relative velocity v . From frame- S , a particle of rest mass m_0 moves with speed w . Its speed w' from frame- S' is given by

$$w' = \frac{w - v}{1 - \frac{vw}{c^2}}$$

The energy of this particle from frame- S is given by

$$E = \frac{m_0c^2}{\sqrt{1 - w^2/c^2}}$$

Similarly, its energy from frame- S' is given by

$$E' = \frac{m_0c^2}{\sqrt{1 - w'^2/c^2}}$$

I can substitute for w' in this equation with what I got earlier and thus reach

$$\begin{aligned} E' &= \frac{m_0c^2}{\sqrt{1 - \frac{(w-v)^2}{c^2(1-vw/c^2)^2}}} \\ &= \frac{E\sqrt{1 - w^2/c^2} \times c^2(1 - vw/c^2)}{\sqrt{(c^2 - vw)^2 - c^2(w - v)^2}} \\ &= \frac{E\sqrt{1 - w^2/c^2} \times c^2(1 - vw/c^2)}{\sqrt{(c^2 - v^2)(c^2 - w^2)}} \\ &= \frac{E(1 - vw/c^2)}{\sqrt{1 - v^2/c^2}} \end{aligned}$$

Technically from this, I can transform the energy of a particle from one frame to another knowing the relative velocity between the frames. But I'm not satisfied. I want it in a form devoid of w or w' since we are already incorporating energy or momentum of the particle from that frame. I can further write E as

$$\begin{aligned} E' &= \gamma[v] \times E \left(1 - \frac{vw}{c^2}\right) \\ &= \gamma \left(E - \frac{(mc^2)vw}{c^2}\right) \\ &= \gamma(E - (mw)v) \end{aligned}$$

I can write mw as the relativistic momentum of the particle (since m here is the relativistic mass and not the rest-mass of the particle) to get

$$\boxed{E' = \gamma(E - vp)}$$

Similarly, the momentum of the above particle from frame- S is given by

$$p = \frac{m_0 w}{\sqrt{1 - w^2/c^2}}$$

and the same from frame- S' is given by

$$p' = \frac{m_0 w'}{\sqrt{1 - w'^2/c^2}}$$

Just as before, I replace w' in this equation with terms in w to get

$$\begin{aligned} p' &= \frac{m_0 \left(\frac{w-v}{1-vw/c^2}\right)}{\sqrt{1 - \frac{(w-v)^2}{c^2(1-vw/c^2)^2}}} \\ &= \frac{m_0 c(w-v)}{\sqrt{c^2(1-vw/c^2)^2 - (w-v)^2}} \\ &= \frac{m_0 c^2(w-v)}{\sqrt{(c^2-v^2)(c^2-w^2)}} \\ &= \frac{m_0(w-v)}{\sqrt{\left(1 - \frac{v^2}{c^2}\right) \left(1 - \frac{w^2}{c^2}\right)}} \end{aligned}$$

Again, I can be satisfied with this, but no! I want a form without w since E and p takes care of that

$$\begin{aligned} p' &= m_0(w-v) \gamma[v] \gamma[w] \\ &= \gamma[v] [mw - mv] \end{aligned}$$

I can write $mw = p$ and $mv = (mc^2)v/c^2 = E/c^2$ and thus

$$\boxed{p' = \gamma \left(p - \frac{Ev}{c^2}\right)}$$

Thus we have got two more transformation laws – One for energy and one for momentum. Remember that the γ in these formulas refer to Lorentz factor for the frames and not for the particle under consideration (ie. $\gamma[v]$ not $\gamma[w]$)

One cannot ignore how the transformation of $E' \iff E$ is exactly similar to the Lorentz transformation of $x' \iff x$ and the transformation of $p' \iff p$ is similar to that of the Lorentz transformation of $t' \iff t$. We can thus write a four-vector representation for energy and momentum (along with the same Lorentz transformation matrix

$$\begin{bmatrix} E'/c \\ p'_x \\ p'_y \\ p'_z \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma v/c & 0 & 0 \\ -\gamma v/c & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} E/c \\ p_x \\ p_y \\ p_z \end{bmatrix}$$

If that is true, then just as how $-(ct)^2 + x^2$ is a Lorentz invariant, so must be $E^2 - (cp)^2$. Let's test that out. Substituting the equations for E and p , I get

$$\begin{aligned} E^2 - (cp)^2 &= \left(\frac{m_0 c^2}{\sqrt{1 - w^2/c^2}} \right)^2 - \left(\frac{cm_0 w}{\sqrt{1 - w^2/c^2}} \right)^2 \\ &= \frac{m_0^2 c^4 - m_0^2 w^2 c^2}{1 - w^2/c^2} \\ &= \frac{m_0 c^4 (1 - w^2/c^2)}{1 - w^2/c^2} = m_0 c^4 \end{aligned}$$

I get a quantity which is independent of any terms of velocity (v or w) and therefore it must be same for $E'^2 - c^2 p'^2$ as well. Thus we get yet another valid invariant in Special Relativity.

$$\boxed{E^2 - c^2 p^2 = E'^2 - c^2 p'^2}$$

15 Conclusion

We began our discussions with a paradox between Galilean Relativity and the speed of light according to Maxwell's Equations. We decided to make a few modifications to Galilean Relativity and started calling it special relativity. We saw how that minor adjustment fixed the issue we were having with maintaining a constant speed of light but lead to an innumerable number of other consequences. Time that was absolute earlier turned out to be not in this new framework, length of bodies and space that we expected to be invariant turned out to be not, we discovered a new set of absolute quantities in this theory that we called invariants, we formulated the mathematics that is required to go from one person's perspective to another, we derived a new more complex looking formula to add and subtract velocities, we saw how a pictorial representations called space-time diagrams come handy in solving problems more efficiently and understand the picture behind what is happening. And Finally we saw how energy and matter are interchangeable entities from the realization that mass too is not absolute... I hope I was able to deliver these ideas well through my unusual (yet inspired from David Griffiths) pedagogical style. Thank you for joining me on this journey through the wonderful world of Special Relativity...

The limitations of Special Relativity and the rise of General Relativity :-

In this bonus section, we'll explore the limitations of Special Relativity, possible work-a-rounds, and discuss an alternative solution. First limitation is, as you might have guessed, the necessity of inertial reference frames. We have always avoided accelerations because it is hard to deal with in special relativity (it is possible to do so by considering infinitesimal time intervals where velocity can be considered constant and with the help of a bit of integral calculus). We also arrive at something called hyperbolic motion through Minkowski (flat) space-time. Hyperbolic because our classical notion of a constant acceleration translates to one that asymptotically approaches c in special relativity. Thus Special Relativity is actually cool even when dealing with accelerated motion as long as the space-time is flat.

The true wall actually appears when we incorporate gravity. To give an example, earlier we saw how a constant force can never accelerate a body past c . But what if this force was proportional to the mass itself like how we derive gravitational acceleration $mg = ma \implies a = g$. Then it seems like this force can very well accelerate the mass beyond c . Also, Newton's Universal law of gravitation assumes gravitation acts instantly across space not respecting the maximum permissible speed of information according to special Relativity. During this time, Einstein had the realization that a person standing inside a closed box accelerating upwards at 9.8m/s^2 in free space experiences a phenomena that is indistinguishable from gravity on earth and similarly free-fall under gravity is indistinguishable from floating in deep space (inertial frame). This is called the equivalence principle.

A photon shot from one wall of this box will hit the opposite wall a bit lower. That is, acceleration seems to bend light from the box's frame. By equivalence principle, this would mean that gravity would bend light too. Another example is that of rotating disk. The circumference of the disk becomes Lorentz contracted (being tangential to the direction of motion at every instant) meanwhile the radius remains same (being perpendicular to the motion at every instant). That would imply that ratio of circumference to radius is no longer 2π . Thus accelerated frames imposes non-euclidean geometry. If accelerated frames force us to abandon Euclidean geometry, and the Equivalence Principle tells us that acceleration is locally indistinguishable from gravity, the conclusion becomes clear – gravity itself must be a manifestation of non-Euclidean, curved spacetime.

This brings us to an outstanding realization. Gravity is no longer an invisible Newtonian force pulling objects across flat space at infinite speed. Instead, mass and energy warp the fabric of spacetime, and objects in free fall are simply following the straightest possible paths—called geodesics—through this curved geometry. John Wheeler said it best “Spacetime tells matter how to move and matter tells spacetime how to curve.” With this conceptual leap, the limitations of Special Relativity dissolve leading to the General Theory of Relativity.